

COUNTEREXAMPLES, SPECTRAL OBSTRUCTIONS, AND DELETION STABILITY FOR WOW-284

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ABSTRACT. WOW-284 asserts that the minimum dual degree of every connected graph of order at least three and girth at least five does not exceed the negative of its least distance eigenvalue. We refute it with exact counterexamples of orders 38, 39, 40, 42, and 50, and develop a structural theory of the failure. For a connected k -regular graph of girth at least five and diameter three, we prove

$$\delta^*(G) + \lambda_{\min}(D(G)) = 2k - 2 - \max_{\theta \neq k}(\theta + 1)^2.$$

Here θ ranges over the nonprincipal adjacency eigenvalues. Consequently every regular strict counterexample has degree at least six and diameter at most four, while diameter four forces degree at least ten. We solve the associated one-variable nonbacktracking linear program exactly, including optimizer rigidity. For regular strict counterexamples of diameter three, the optimizer yields a positive-semidefinite slack matrix whose integral excess gives the stronger bound

$$|V(G)| \leq \left\lfloor \frac{3(k+2)^2(k^2+3)}{18k+41} \right\rfloor;$$

this follows from a three-to-one quantization theorem for the integral excess. The slack matrix's principal minors also recover local cycle constraints. In particular, regular degree-six counterexamples have order at most 50, and at the degree-six, order-50 boundary the associated signed complement is necessarily disconnected. We determine the distance spectra of one- and two-vertex punctures of Moore graphs and establish a uniform deletion-stability bound: every deletion of at most five vertices from the Hoffman–Singleton graph remains a strict counterexample, whereas an explicit six-vertex deletion does not. All theorem-level computations use exact arithmetic. Lean 4.31 kernel-checks the explicit 50-vertex Hoffman–Singleton counterexample at graph level, finite spectral certificates at orders 38, 39, 40, 42, and the analytic LP optimum and rigidity for every integer $k \geq 4$.

1. INTRODUCTION

Let G be a finite simple connected graph of order $n = |V(G)| \geq 2$. Let $A = A(G)$ and $D = D(G)$ be its adjacency and distance matrices, let I and J denote the identity and all-ones matrices of order n , and let $\mathbf{1}$ be the all-ones vector. Write $d(v)$ for the degree of v , $N(v)$ for its open neighbourhood, and, for $S \subseteq V(G)$, let $G - S$ denote the subgraph induced by $V(G) \setminus S$. Define

$$d^*(v) = \frac{1}{d(v)} \sum_{u \in N(v)} d(u), \quad \delta^*(G) = \min_{v \in V(G)} d^*(v).$$

Thus $D = (d_G(u, v))_{u, v \in V(G)}$. Its eigenvalues are ordered as

$$\partial_1(G) \geq \dots \geq \partial_n(G), \quad \partial_n(G) = \lambda_{\min}(D(G)).$$

Aouchiche and Hansen record the following Graffiti conjecture as Conjecture 7.16 and attribute it to Fajtlowicz's 1998 *Written on the Wall* report [1] (see also [2, Conjecture 7.16]).

Conjecture (WOW-284). *If G has order at least three and girth at least five, then*

$$\delta^*(G) \leq -\lambda_{\min}(D(G)).$$

For graphs in the domain of the conjecture, put

$$\Phi(G) = \delta^*(G) + \lambda_{\min}(D(G)).$$

Thus G is a strict counterexample precisely when $\Phi(G) > 0$.

The initial disproof is short. A degree- k Moore graph of diameter two has

$$A^2 = (k-1)I - A + J, \quad D = 2J - 2I - A,$$

and hence

$$\delta^*(G) = k, \quad \lambda_{\min}(D) = -\frac{3 + \sqrt{4k-3}}{2}.$$

The conjecture holds on these graphs exactly for $k \leq 3$, with equality at $k = 3$, and fails for every realizable $k > 3$. The degree-seven Hoffman–Singleton graph therefore gives a gap of three.

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The purpose of this paper is not merely to list descendants of this graph. It addresses three structural questions.

- (1) Which spectral mechanism governs regular counterexamples?
- (2) How restrictive are the degree, diameter, and order conditions?
- (3) How stable is the counterexample property under deletion?

Our main conclusions are as follows.

- For connected regular graphs of girth at least five and diameter three, WOW-284 is equivalent to a two-sided adjacency-spectrum window centred at -1 ; see Theorem 3.1.
- Every regular strict counterexample has degree at least six and diameter at most four. A diameter-four example, if one exists, has degree at least ten; see Theorems 4.2, 4.3, and 4.4.
- The standard one-variable nonbacktracking linear-programming hierarchy has exact ceiling

$$B_k = \frac{(k+2)(k^2+3)}{6},$$

with a unique optimizer up to positive scaling; see Theorem 5.3.

- For regular strict counterexamples of diameter three, the optimizer defines a positive-semidefinite slack matrix. Writing

$$r = 2(k+2)^2(k^2+3) - (12k+27)n,$$

its integral excess satisfies $r > 0$ and $n \leq 3r$, giving

$$n \leq \left\lfloor \frac{3(k+2)^2(k^2+3)}{18k+41} \right\rfloor;$$

see Theorems 5.4 and 6.1. Equality in the unrounded inequality is arithmetically rigid:

$$(k, n, r) = (103, 185220, 61740);$$

see Corollary 6.2. The same slack matrix's 2×2 minors yield a general cycle-divisibility sieve and the local degree-six, order-51 contradiction

$$5N_5 = 153 \cdot 11 = 1683,$$

where N_5 denotes the number of 5-cycles. The global theorem gives the degree-7, 8, 9 order windows 74, 108, 150; see Corollary 6.7.

- At the unresolved degree-six order-50 boundary, the integral signed-complement Gram matrix has rank at least 30, and its underlying signed graph is disconnected; see Proposition 6.8 and Theorem 6.11.
- One-vertex, adjacent-pair, and nonadjacent-pair deletions of Moore graphs admit exact invariant-subspace decompositions for their recomputed distance matrices; see Section 7.
- Every deletion of at most five vertices from the Hoffman–Singleton graph remains a strict counterexample, and this universal radius is sharp; see Theorem 8.3.

The proofs form two complementary hierarchies. In the obstruction direction, the distance-polynomial identity converts WOW-284 into a shifted adjacency window; scalar trace moments give the one-point LP ceiling; integrality of the optimal slack matrix strengthens the order bound; graph realizability and small Gram minors quantize the remaining excess into the three-to-one bound; edge-local 2×2 minors convert the same certificate into cycle counts; and 3×3 minors impose the two-path constraints at order fifty, where a signed-root representation forces a nontrivial component decomposition. In the stability direction, deleting vertices from a Moore graph produces an incidence-Gram correction to the distance matrix and a configuration-sensitive perturbation bound. Invariant-subspace decompositions then give exact puncture spectra, while orbitwise positive-definiteness certificates determine the sharp Hoffman–Singleton deletion radius.

The distance-polynomial viewpoint is established for minimal cages and distance-polynomial graphs [3, 4]. Nonbacktracking linear-programming bounds are due to Nozaki [5]; related spectral-Moore work appears in [6]. Our contribution is the specialization to the two-sided WOW window, the exact optimum for the admissible LP class of Section 5, the integral optimal-slack hierarchy, the edge-local cycle certificate, and the deletion theory developed below. We give an exact refutation of WOW-284 and a Lean 4.31 graph-level formalization of the 50-vertex certificate. We do not claim that order 38 is minimum or that the constructions classify all counterexamples.

The analytic arguments are proved in the text. Precisely specified finite classifications and matrix certificates are treated as computer-assisted proof components. Their exact reproducibility materials are archived with the accompanying release.

2. LOCAL GROWTH, MOORE GRAPHS, AND EXPLICIT COUNTEREXAMPLES

2.1. Dual degree as radius-two growth. For $v \in V(G)$, write $\Gamma_i(v)$ for the distance- i sphere and $B_2(v) = \Gamma_0(v) \cup \Gamma_1(v) \cup \Gamma_2(v)$.

Proposition 2.1 (Second-degree identity). *If G contains no triangle and no 4-cycle, then*

$$|B_2(v)| = 1 + \sum_{u \in N(v)} d(u), \quad d^*(v) = \frac{|B_2(v)| - 1}{d(v)}.$$

Proof. For distinct neighbours u, w of v , the sets $N(u) \setminus \{v\}$ and $N(w) \setminus \{v\}$ are disjoint; an intersection would form a 4-cycle, and a member in $N(v)$ would form a triangle. These sets partition $\Gamma_2(v)$, so

$$|\Gamma_2(v)| = \sum_{u \in N(v)} (d(u) - 1).$$

Adding the centre and first sphere proves the formula. This is the normalized form recorded in Backelin's Lemma 2.1 [7]. $\square$

2.2. The Moore threshold.

Theorem 2.2. *Let M be a degree- k Moore graph of diameter two, $k \geq 2$. Then*

$$|V(M)| = k^2 + 1, \quad g(M) = 5, \quad \delta^*(M) = k,$$

$$\lambda_{\min}(D(M)) = -\frac{3 + \sqrt{4k - 3}}{2},$$

and

$$\Phi(M) = k - \frac{3 + \sqrt{4k - 3}}{2}.$$

Thus M satisfies WOW-284 exactly for $k \leq 3$, with equality exactly at $k = 3$.

Proof. The Moore bound is attained, so adjacent vertices have no common neighbour and nonadjacent vertices have exactly one. Given an edge uv , choose $x \in N(u) \setminus \{v\}$ and $y \in N(v) \setminus \{u\}$. The vertices x, y are nonadjacent, since an edge would create a four-cycle, and their unique common neighbour completes a five-cycle through uv . Therefore

$$A^2 = (k - 1)I - A + J.$$

On $\mathbf{1}^\perp$, the nonprincipal adjacency eigenvalues are the roots

$$r, s = \frac{-1 \pm \sqrt{4k - 3}}{2}.$$

If m_r, m_s are their multiplicities, then

$$m_r + m_s = k^2, \quad k + m_r r + m_s s = 0.$$

Consequently

$$m_r = \frac{k(k\sqrt{4k-3} + k - 2)}{2\sqrt{4k-3}}, \quad m_s = \frac{k(k\sqrt{4k-3} - k + 2)}{2\sqrt{4k-3}},$$

and both roots occur. Every nonedge has distance two, hence $D = 2J - 2I - A$. The least distance eigenvalue is $-2 - r = -(3 + \sqrt{4k - 3})/2$. Regularity gives $\delta^* = k$, and

$$(2k - 3)^2 - (4k - 3) = 4(k - 1)(k - 3)$$

gives the threshold. The exact scalar and finite checks are independently repeated by `scripts/verify_regular_score_calculus.py`. $\square$

2.3. A coordinate Hoffman–Singleton certificate. All subscripts below lie in $\mathbb{F}_5 = \mathbb{Z}/5\mathbb{Z}$. Let

$$V(M) = \{P_{i,j} : i, j \in \mathbb{F}_5\} \dot{\cup} \{Q_{k,\ell} : k, \ell \in \mathbb{F}_5\},$$

with edges

$$P_{i,j} \sim P_{i,j \pm 1},$$

$$Q_{k,\ell} \sim Q_{k,\ell \pm 2},$$

$$P_{i,j} \sim Q_{k,ik+j}.$$

This is Hafner's affine-coordinate form of the Hoffman–Singleton graph after a minor reindexing [8].

Proposition 2.3. *The coordinate construction is a simple connected 7-regular graph on 50 vertices. Adjacent pairs have no common neighbour and nonadjacent pairs have exactly one. Consequently it has girth five, diameter two, and*

$$\text{Spec } D(M) = \{91^{(1)}, 1^{(21)}, (-4)^{(28)}\}.$$

In particular, $\delta^(M) = 7$ and $\Phi(M) = 3$.*

Proof. The neighbourhoods are

$$N(P_{i,j}) = \{P_{i,j-1}, P_{i,j+1}\} \cup \{Q_{k,ik+j} : k \in \mathbb{F}_5\},$$

$$N(Q_{k,\ell}) = \{Q_{k,\ell-2}, Q_{k,\ell+2}\} \cup \{P_{i,\ell-ik} : i \in \mathbb{F}_5\}.$$

They have seven distinct entries. For two P -vertices in the same layer, the 5-cycle gives no common neighbour when they are adjacent and exactly one when they are nonadjacent; a common Q -neighbour would force their second coordinates to agree. In distinct P -layers, a common Q -neighbour is determined uniquely by $(i - i')k = j' - j$. The Q -cases are identical, with the same-layer 5-cycle generated by steps ± 2 and, in distinct layers, a unique common P -neighbour. For a cross pair $P_{i,j}, Q_{k,\ell}$, put $r = \ell - (ik + j)$. The pair is adjacent for $r = 0$, has one common P -neighbour for $r \in \{\pm 1\}$, and one common Q -neighbour for $r \in \{\pm 2\}$. The five residues are exhausted. The claimed geometry now follows from Theorem 2.2. For degree 7, the multiplicity equations in its proof give adjacency multiplicities 28 at 2 and 21 at -3 , and hence the displayed distance spectrum. The exhaustive pair certificate, integer BFS distances, characteristic polynomial, and exact positive-definiteness check are in `scripts/verify_exact.py`. $\square$

2.4. Smaller exact counterexamples. Let

$$\mathcal{P} = \{P_{0,j}, Q_{0,j} : j \in \mathbb{F}_5\}.$$

The induced graph $M[\mathcal{P}]$ is a Petersen graph. Put

$$R = M - \mathcal{P}, \quad H_{39} = R - P_{1,0}, \quad H_{38} = R - \{P_{1,0}, P_{1,1}\},$$

and let X_{42} be the second subconstituent of $P_{0,0}$, namely the graph induced by the vertices at distance two from it.

Theorem 2.4. *The following are strict counterexamples.*

G	$ V(G) $	$\delta^*(G)$	$\lambda_{\min}(D(G))$
H_{38}	38	$17/3$	$-3 - \sqrt{7}$
H_{39}	39	$35/6$	$> -35/6$
R	40	6	-5
X_{42}	42	6	-5
M	50	7	-4

The entry for H_{39} records an exact strict lower bound obtained from positive definiteness of $6D + 35I$; it is not a decimal approximation to the least eigenvalue. Moreover, all 40 labelled singleton deletions of R and all 120 labelled deletions of the endpoints of an edge of R are strict counterexamples; within each family, the distance characteristic polynomial is constant.

Proof. The graph R is the classical $(6, 5)$ -cage of O’Keefe and Wong [9, 10]; its realization as a Petersen deletion of the Hoffman–Singleton graph also appears in [11, pp. 262–263]. The Moore block identity gives

$$\text{Spec } A(R) = \{6^{(1)}, 2^{(18)}, 1^{(4)}, (-2)^{(5)}, (-3)^{(12)}\},$$

and Theorem 3.1 below maps this to

$$\text{Spec } D(R) = \{75^{(1)}, 3^{(5)}, 0^{(16)}, (-5)^{(18)}\}.$$

The second-subconstituent calculation gives

$$\text{Spec } D(X_{42}) = \{81^{(1)}, 4^{(6)}, 0^{(14)}, (-5)^{(21)}\}.$$

The classical second-subconstituent identification and adjacency spectrum are recorded in [12, Table 3, p. 265]. For H_{38} , a direct degree count gives $\delta^* = 17/3$, while the factor $x^2 + 6x + 2$, together with an exact Sturm isolation, gives the least root $-3 - \sqrt{7}$. For H_{39} , the exact matrix $6D + 35I$ is positive definite. All graph, girth, distance, dual-degree, Sturm, and rational LDL^T certificates are checked by `scripts/verify_extended.py`. More explicitly, for every $v \in V(R)$,

$$\begin{aligned} \det(xI - D(R - v)) &= P_{39}(x) := x^9(x + 5)^{12}(x^2 + 6x + 3) \\ &\quad \cdot (x^3 + 3x^2 - 15x - 7)^2(x^3 + 3x^2 - 15x - 3)^2 \\ &\quad \cdot (x^4 - 78x^3 + 303x^2 - 70x - 450), \end{aligned}$$

whereas for every $uv \in E(R)$,

$$\begin{aligned} \det(xI - D(R - \{u, v\})) &= P_{38}(x) := x^4(x-2)(x+5)^8(x^2+6x+2)^2 \\ &\quad \cdot (x^3+3x^2-15x-3) \\ &\quad \cdot (x^4+5x^3-7x^2-23x-6) \\ &\quad \cdot (x^5+9x^4+7x^3-77x^2-54x-4) \\ &\quad \cdot (x^9-67x^8-404x^7+1772x^6+7205x^5 \\ &\quad - 18489x^4-17018x^3+20288x^2+16824x+1680). \end{aligned}$$

The labelled deletion families are exhausted by `scripts/verify_descendant_families.py`. No transitivity or numerical root ordering is used. Fixed graph6, adjacency-list, and edge-list records are provided in `data/graphs/`. $\square$

Theorem 2.5 (Moore second subconstituents). *Let M be a degree- K Moore graph of diameter two, $K \geq 3$, and let X be the graph induced by $\Gamma_2(v)$ for a fixed vertex v . Then X has order $K(K-1)$, degree $K-1$, girth at least five, diameter three, and*

$$\lambda_{\min}(D(X)) = -\frac{5 + \sqrt{4K-3}}{2}.$$

Consequently

$$\Phi(X) = K-1 - \frac{5 + \sqrt{4K-3}}{2},$$

which is positive exactly for integers $K \geq 6$.

Proof. Put $m = K(K-1)$. The Moore property gives $|\Gamma_2(v)| = m$. Each vertex of $\Gamma_2(v)$ has a unique neighbour in $N(v)$, while each vertex of $N(v)$ has $K-1$ neighbours in $\Gamma_2(v)$. Hence X has order m , is $(K-1)$ -regular, and inherits girth at least five from M . We prove below that its diameter is three.

Relative to $\{v\} \sqcup N(v) \sqcup \Gamma_2(v)$, let $C \in \{0, 1\}^{K \times m}$ be the incidence block from $N(v)$ to $\Gamma_2(v)$, and let $B \in \{0, 1\}^{m \times m}$ be the adjacency matrix of X . Write $J_{a,b}$ for the $a \times b$ all-ones matrix and $J_a = J_{a,a}$. Then

$$C\mathbf{1}_m = (K-1)\mathbf{1}_K, \quad C^\top \mathbf{1}_K = \mathbf{1}_m.$$

The Moore identity, read in the corresponding blocks, gives

$$CC^\top = (K-1)I_K, \quad CB = J_{K,m} - C,$$

and

$$C^\top C + B^2 = (K-1)I_m - B + J_m.$$

Since $CC^\top = (K-1)I_K$, the map $C^\top$ is injective and

$$\mathbb{R}^m = \langle \mathbf{1}_m \rangle \perp C^\top(\mathbf{1}_K^\perp) \perp \ker C.$$

Moreover, $B\mathbf{1}_m = (K-1)\mathbf{1}_m$, and transposing the identity for CB shows that B acts as $-I$ on $C^\top(\mathbf{1}_K^\perp)$. Also, $\ker C \subseteq \mathbf{1}_m^\perp$; the same identity shows that $\ker C$ is B -invariant, and the final block identity gives

$$(B^2 + B - (K-1)I_m)|_{\ker C} = 0.$$

Thus the remaining eigenvalues are among

$$r, s = \frac{-1 \pm \sqrt{4K-3}}{2}.$$

Finally,

$$\dim \ker C = m - K = K(K-2), \quad \text{tr}(B|_{\ker C}) = 0,$$

because the eigenvalue $K-1$ cancels the $K-1$ copies of -1 in $\text{tr } B = 0$. Hence the multiplicities p, q of r, s satisfy

$$p + q = K(K-2), \quad pr + qs = 0,$$

so

$$p = \frac{K(K-2)(1 + \sqrt{4K-3})}{2\sqrt{4K-3}}, \quad q = \frac{K(K-2)(\sqrt{4K-3} - 1)}{2\sqrt{4K-3}}.$$

Both roots therefore occur. It remains to identify the diameter of X . If the unique common neighbour in M of two nonadjacent vertices $x, y \in X$ lies in X , their distance in X is two. Otherwise it is their common parent in $N(v)$. Choose $b \in N_X(x)$. The parent of b differs from the common parent of x, y , since otherwise those three vertices would form a triangle. Moreover $b \not\sim y$, and the unique common neighbour c of b, y belongs to X : it cannot be v , and it cannot lie in $N(v)$, since b and y have different parents there. Thus $x - b - c - y$ is a path in X . Each parent has $K-1 \geq 2$ children, and pairs of distinct children have no length-two path in X . Thus X has diameter three

and Theorem 3.1 applies. Among the nonprincipal adjacency eigenvalues, $(-1 + \sqrt{4K - 3})/2$ maximizes $|\theta + 1|$; substitution gives

$$\lambda_{\min}(D(X)) = -\frac{5 + \sqrt{4K - 3}}{2}.$$

For $K = 3, 4, 5$, direct comparison gives a nonpositive score. For $K \geq 6$, both sides of the relevant comparison are positive, and the threshold reduces to

$$(2K - 7)^2 - (4K - 3) = 4(K^2 - 8K + 13) > 0$$

because $K > 4 + \sqrt{3}$. The finite $K = 7$ instance is checked by `scripts/verify_wow284_38_40_42.py`. $\square$

3. THE REGULAR SCORE CALCULUS

Theorem 3.1 (Diameter-three score formula). *Let G be connected, k -regular, of girth at least five and diameter three, with adjacency matrix A and order n . Then*

$$D = 3J + (k - 3)I - 2A - A^2,$$

$$D + kI = 3J + (2k - 2)I - (A + I)^2.$$

The principal distance eigenvalue is $3n - k^2 - k - 3$, and a nonprincipal adjacency eigenvalue θ gives the distance eigenvalue

$$\mu(\theta) = k - 2 - (\theta + 1)^2.$$

Consequently

$$\Phi(G) = 2k - 2 - \max_{\theta \neq k} (\theta + 1)^2.$$

Thus G is a strict counterexample exactly when every nonprincipal adjacency eigenvalue θ satisfies

$$|\theta + 1| < \sqrt{2k - 2}$$

Proof. Girth at least five gives the distance-two matrix $A_2 = A^2 - kI$, and diameter three gives $A_3 = J - I - A - A_2$. Substitute in $D = A + 2A_2 + 3A_3$. On $\mathbf{1}^\perp$, J vanishes, and regularity gives $\delta^* = k$. The principal distance eigenvalue is positive and is the Perron root because every off-diagonal entry of D is positive. The resulting spectral-transfer and score calculations for the examples used below are independently checked by `scripts/verify_regular_score_calculus.py`. $\square$

Corollary 3.2 (Bipartite obstruction). *A connected k -regular bipartite graph of girth at least five and diameter three is not a strict counterexample for $k \geq 3$.*

Proof. The nonprincipal eigenvalue $-k$ satisfies

$$(-k + 1)^2 - (2k - 2) = (k - 1)(k - 3) \geq 0 \quad (k \geq 3).$$

Thus the required strict inequality fails, with equality exactly at $k = 3$. $\square$

Proposition 3.3 (Higher-diameter transfer). *Let G be connected and k -regular, with diameter d and girth at least $2d - 1$. Let A_i denote the distance- i matrix, and define*

$$F_0 = 1, \quad F_1 = x, \quad F_2 = x^2 - k, \quad F_i = xF_{i-1} - (k - 1)F_{i-2} \quad (i \geq 3).$$

Then $A_i = F_i(A)$ for $0 \leq i \leq d - 1$, and

$$D = dJ + q_d(A), \quad q_d(x) = \sum_{i=0}^{d-1} (i - d)F_i(x).$$

In particular,

$$\begin{aligned} q_3(x) &= k - 3 - 2x - x^2, \\ q_4(x) &= -x^3 - 2x^2 + (2k - 4)x + 2k - 4. \end{aligned}$$

Proof. Up to length $d - 1$, the girth condition makes nonbacktracking walks between two vertices unique exactly when their length is the graph distance. Hence the distance- i matrices are the nonbacktracking polynomials in A ; summing $D = \sum_{i=0}^d iA_i$ and eliminating A_d with $J = \sum_{i=0}^d A_i$ proves the formula. This lies within the established distance-polynomial framework [3, 4]. $\square$

4. DEGREE AND DIAMETER OBSTRUCTIONS

Lemma 4.1. *For every connected graph,*

$$\lambda_{\min}(D(G)) \leq -\text{diam}(G).$$

Proof. For a diametral pair u, v , the Rayleigh quotient of $e_u - e_v$ is $-d_G(u, v)$. $\square$

Theorem 4.2. *Every connected regular strict counterexample to WOW-284 has degree at least six.*

Proof. We use the LP ceiling proved independently in Theorem 5.3; that theorem does not depend on the present degree reduction. Let the degree be k . Lemma 4.1 and strictness give $\text{diam}(G) < k$. If $k \leq 2$, then $|V(G)| \geq 3$, while the girth hypothesis excludes completeness; hence $\text{diam}(G) \geq 2 \geq k$, a contradiction. For $k = 3$, the radius-two lower bound and $\text{diam}(G) \leq 2$ force equality in the Moore bound. Hence G is a degree-three Moore graph, and Theorem 2.2 gives $\Phi(G) = 0$, not $\Phi(G) > 0$.

For $k = 4$, diameter two would require a degree-four Moore graph, whose adjacency multiplicities are nonintegral. In diameter three, the exact LP bound of Theorem 5.3 gives $n < 19$, whereas the radius-two ball has 17 vertices and diameter three requires at least one more. Hence $n = 18$. The distance-three matrix is then a perfect matching. Its -1 -eigenspace $W = \ker(A_3 + I)$ is a nine-dimensional rational subspace of $\mathbf{1}^\perp$. Since A commutes with A_3 , the space W is A -invariant, and

$$A_3 = J + 3I - A - A^2 \implies (A^2 + A - 4I)|_W = 0.$$

The polynomial $x^2 + x - 4$ is irreducible over $\mathbb{Q}$, so a rational space on which it annihilates an operator has even dimension, a contradiction.

For $k = 5$, diameter two again fails the Moore multiplicity condition. A diametral geodesic in diameter four yields the principal submatrix $D(P_5)$, whose factor $x^2 + 6x + 4$ supplies the eigenvalue $-3 - \sqrt{5} < -5$; Cauchy interlacing excludes this case. In diameter three, Meringer's enumeration [13, p. 142] and Theorem 5.3 leave $n \in \{30, 31, 32\}$. Since $5n = 2|E(G)|$, one has $n \equiv 0 \pmod{2}$, so $n \in \{30, 32\}$. At $n = 32$, the distance layers about a vertex have sizes 1, 5, 20, 6. Write a for the average internal degree of the distance-two layer. Each of its vertices has one neighbour in the first layer, so the number of edges from the second to the third layer is $20(4 - a) \leq 6 \cdot 5$; hence $a \geq 5/2$. On normalized layer indicators, the symmetric adjacency compression is

$$Q(a) = \begin{pmatrix} 0 & \sqrt{5} & 0 & 0 \\ \sqrt{5} & 0 & 2 & 0 \\ 0 & 2 & a & (4-a)\sqrt{10/3} \\ 0 & 0 & (4-a)\sqrt{10/3} & 5 - \frac{10}{3}(4-a) \end{pmatrix}.$$

The derivative of its only a -dependent block is

$$\begin{pmatrix} 1 & -\sqrt{10/3} \\ -\sqrt{10/3} & 10/3 \end{pmatrix} = \begin{pmatrix} 1 \\ -\sqrt{10/3} \end{pmatrix} \begin{pmatrix} 1 & -\sqrt{10/3} \end{pmatrix} \succeq 0.$$

Thus every ordered eigenvalue of $Q(a)$ is nondecreasing in a . Write $5 = \mu_1(a) \geq \mu_2(a) \geq \mu_3(a) \geq \mu_4(a)$. At the smallest feasible value,

$$\chi_{Q(5/2)}(x) = \frac{1}{4}(x-5)p_{5,6}(x), \quad p_{5,6}(x) = 4x^3 + 10x^2 - 16x - 30, \quad p_{5,6}(11/6) = -29/27 < 0.$$

Since $p_{5,6}(11/6) < 0$ and $p_{5,6}(x) \rightarrow +\infty$ as $x \rightarrow +\infty$, its largest root satisfies $\mu_2(5/2) > 11/6 > -1 + \sqrt{8}$. Monotonicity and Poincaré separation give

$$\theta_2(A) \geq \mu_2(a) \geq \mu_2(5/2),$$

contradicting the necessary bound $\theta_2(A) < -1 + \sqrt{8}$. At $n = 30$, Meringer's isomorph-free enumeration leaves exactly four $(5, 5)$ -cages [13, p. 142]; each fixed record has an exact distance eigenvalue at most -5 . The accompanying release contains an independent exact audit of the complete case split and the four fixed graph6 records. $\square$

Theorem 4.3 (Endpoint-neighbourhood obstruction). *Let G be any connected finite simple graph, and let u, v be vertices at distance $\ell = d_G(u, v) \geq 5$. Put $p = d(u)$ and $q = d(v)$. Then*

$$\lambda_{\min}(D(G)) \leq p + q - 2 - \sqrt{(p - q)^2 + pq(\ell - 2)^2}.$$

If δ is the ordinary minimum degree, then

$$\lambda_{\min}(D(G)) \leq -\delta(\ell - 4) - 2.$$

Consequently every strict WOW-284 counterexample satisfies

$$\Delta > \delta(\ell - 4) + 2,$$

where Δ is the ordinary maximum degree. In particular, every regular strict counterexample has diameter at most four.

Proof. The two endpoint neighbourhoods are disjoint. Give weight $a > 0$ to $N(u)$, weight $-b < 0$ to $N(v)$, and zero elsewhere. Within one neighbourhood distances are at most two; between the two neighbourhoods they are at least $\ell - 2$. Since the cross products are negative, the Rayleigh quotient is at most that of

$$\begin{pmatrix} 2(p-1) & -(\ell-2)\sqrt{pq} \\ -(\ell-2)\sqrt{pq} & 2(q-1) \end{pmatrix}.$$

Its least eigenvalue is the first displayed bound, and its least eigenvector can be chosen with both coordinates positive. Write $p = \delta + \alpha$, $q = \delta + \beta$, and $t = \ell - 2$. The identity

$$(p-q)^2 + pqt^2 - (p+q+\delta(t-2))^2 = (t-2)\{\delta t(\alpha+\beta) + (t+2)\alpha\beta\}$$

is nonnegative. Since $p+q+\delta(t-2) > 0$, comparison of the nonnegative square roots gives the second bound. Finally $\delta^*(G) \leq \Delta$. The sign choice, radical comparison, and integer rounding are independently audited by `scripts/verify_proof_audit_10_endpoint_diameter.py`. $\square$

Theorem 4.4 (Diameter four). *Let G be connected, k -regular, of girth at least five and diameter four. Then*

$$\lambda_{\min}(D(G)) \leq -\frac{7 + \sqrt{16k+1}}{2}.$$

Consequently no such graph of degree $2 \leq k \leq 9$ is a strict counterexample.

Proof. Choose u, v at distance four and put $U = N(u)$, $V = N(v)$. For fixed $a \in U$, every $b \in V$ with $d_G(a, b) = 2$ has a common neighbour in $N(a) \setminus \{u\}$. Distinct such vertices b, b' require distinct common neighbours, since otherwise $b - c - b' - v - b$ is a 4-cycle. Thus at most $k - 1$ vertices of V are at distance two from each a . No pair in $U \times V$ is adjacent, and every remaining pair has distance at least three. Therefore

$$\sum_{a \in U, b \in V} d_G(a, b) \geq 2k^2 + k.$$

For $\alpha, \beta > 0$, assign weights $\alpha, \beta, -\alpha, -\beta$ to u, U, v, V , respectively. Counting unordered pairs and then doubling gives

$$\frac{x^T D(G) x}{x^T x} \leq \frac{-4\alpha^2 - 4k\alpha\beta - 3k\beta^2}{\alpha^2 + k\beta^2}.$$

After setting $y_1 = \alpha$ and $y_2 = \sqrt{k}\beta$, the right-hand side is the Rayleigh quotient of

$$\begin{pmatrix} -4 & -2\sqrt{k} \\ -2\sqrt{k} & -3 \end{pmatrix},$$

whose least eigenvalue is the displayed value and has a positive-coordinate minimizer. The strict comparison with $-k$ holds for $2 \leq k \leq 9$. Every orientation factor, cross-distance sign, and endpoint comparison is independently checked by `scripts/verify_proof_audit_11_diameter_four.py`. $\square$

Corollary 4.5 (Regular trichotomy). *Every regular strict counterexample has one of the following forms:*

- (1) diameter two, hence a Moore graph;
- (2) diameter three, with $|\theta + 1| < \sqrt{2k-2}$ for every nonprincipal adjacency eigenvalue θ ;
- (3) diameter four, with degree at least ten.

There are no regular strict counterexamples of diameter at least five.

5. MOMENT BOUNDS AND THE EXACT LP CEILING

Let G satisfy the hypotheses of Theorem 3.1, and write its nonprincipal adjacency eigenvalues as θ_i . Put $y_i = \theta_i + 1$.

Proposition 5.1 (Fourth-moment identity). *One has*

$$\sum_{i=1}^{n-1} (2k - 2 - y_i^2)(y_i + 1)^2 = (k+2)((k+2)(k^2+3) - 6n).$$

Every strict counterexample therefore satisfies

$$n < B_k := \frac{(k+2)(k^2+3)}{6}.$$

Proof. Use

$$\operatorname{tr} A = \operatorname{tr} A^3 = 0, \quad \operatorname{tr} A^2 = nk, \quad \operatorname{tr} A^4 = nk(2k - 1),$$

remove the principal eigenvalue, and expand both sides. In a strict counterexample, each factor $2k - 2 - y_i^2$ is positive. The sum cannot vanish: otherwise every nonprincipal adjacency eigenvalue would equal -2 , and $\operatorname{tr} A = 0$ would give $k - 2(n - 1) = 0$, or $n = (k + 2)/2$, incompatible with the elementary bound $n \geq k + 1$ for a simple k -regular graph. The identity is checked symbolically by `scripts/verify_degree_six_gate.py` and independently within `scripts/verify_proof_audit_02_two_sided_lp.py`. $\square$

The preceding bound is optimal within the one-variable nonbacktracking admissible class defined below.

Definition 5.2. Let $F_i = F_i^{(k)}$ be the nonbacktracking polynomials from Proposition 3.3, and set

$$I_k = [-1 - \sqrt{2k - 2}, -1 + \sqrt{2k - 2}].$$

A finite polynomial $f = \sum_i f_i F_i$ is *admissible* if

$$f_0 > 0, \quad f_i \geq 0 \quad (i \geq 5), \quad f(x) \leq 0 \quad (x \in I_k).$$

For a k -regular graph of girth at least five whose nonprincipal spectrum lies in I_k , one has $\operatorname{tr} F_i(A) = 0$ for $1 \leq i \leq 4$, while $\operatorname{tr} F_i(A) \geq 0$ for $i \geq 5$, since these traces count closed nonbacktracking walks. The coefficient conditions therefore give $nf_0 \leq \operatorname{tr} f(A)$. On the other hand, $f(\theta) \leq 0$ for every nonprincipal eigenvalue, so $\operatorname{tr} f(A) \leq f(k)$. Thus

$$nf_0 \leq \operatorname{tr} f(A) \leq f(k), \quad n \leq \frac{f(k)}{f_0}.$$

Theorem 5.3 (Exact LP ceiling and rigidity). *For every integer $k \geq 4$ and every admissible f ,*

$$\boxed{\frac{f(k)}{f_0} \geq B_k = \frac{(k + 2)(k^2 + 3)}{6}}.$$

Equality holds if and only if f is a positive scalar multiple of

$$\boxed{f_*(x) = \frac{(x + 2)^2(x^2 + 2x - (2k - 3))}{6(k + 2)}}.$$

Thus increasing the polynomial degree cannot improve this one-variable LP bound. Consequently, any connected k -regular graph of girth at least five whose nonprincipal spectrum lies in the interior of I_k satisfies $n < B_k$.

Proof. The primal expansion is

$$\begin{aligned} 6(k + 2)f_*(x) &= 6(k + 2)F_0(x) + 2(2k + 7)F_1(x) \\ &\quad + (k + 13)F_2(x) + 6F_3(x) + F_4(x). \end{aligned}$$

On I_k , the factor $(x + 1)^2 - (2k - 2)$ is nonpositive, while $(x + 2)^2 \geq 0$; hence f_* is admissible and $f_*(k) = B_k$.

For the dual certificate, put $\Delta = \sqrt{2k - 2}$, $\xi_{\pm} = -1 \pm \Delta$, and $\xi_0 = -2$. Define

$$\begin{aligned} w_- &= \frac{k(k + 2)(2k^2 - 6 - 3(k - 1)\Delta)}{24(2k - 3)}, \\ w_0 &= \frac{k(k - 1)(k^2 + 3)}{6(2k - 3)}, \\ w_+ &= \frac{k(k + 2)(2k^2 - 6 + 3(k - 1)\Delta)}{24(2k - 3)}. \end{aligned}$$

$w_-, w_0, w_+ > 0$. For the only nontrivial inequality, this follows from

$$(2k^2 - 6)^2 - 18(k - 1)^3 = 2(k - 3)(2k - 3)(k^2 + 3) > 0.$$

The measure $\mu = w_- \delta_{\xi_-} + w_0 \delta_{\xi_0} + w_+ \delta_{\xi_+}$ satisfies

$$\mu(1) = B_k - 1, \quad \mu(F_i) = -F_i(k) \quad (1 \leq i \leq 4).$$

For $i \geq 5$, the slack $a_i = \mu(F_i) + F_i(k)$ is strictly positive. For $5 \leq i \leq 9$, exact calculation gives, after removing the common positive factor $k(k - 1)(k + 2)(k^2 + 3)/6$, respectively,

$$\begin{aligned} &2, \quad 5k - 13, \quad 2(3k^2 - 17k + 25), \\ &6k^3 - 47k^2 + 139k - 150, \quad 2(3k^4 - 27k^3 + 106k^2 - 219k + 194), \end{aligned}$$

all positive for $k \geq 4$; for the nontrivial residual factors this follows after writing $k = m + 4$, when all coefficients are nonnegative and the constant terms are positive. For $i \geq 10$, put $r = k - 1 \geq 3$. The support lies in $[-2\sqrt{r}, 2\sqrt{r}]$, because $1 + \sqrt{2r} \leq 2\sqrt{r}$. For $|z| \leq 1$, the recurrence gives

$$F_i(2\sqrt{r}z) = r^{i/2}U_i(z) - r^{(i-2)/2}U_{i-2}(z),$$

where U_j is the Chebyshev polynomial of the second kind. Using $|U_j(z)| \leq j + 1$ yields

$$\frac{|\mu(F_i)|}{F_i(k)} \leq \frac{2i+1}{3} 3^{3-i/2}.$$

At $i = 10$ the right-hand side is $7/9$, and it decreases thereafter because $3(2i+1)^2 - (2i+3)^2 = 8i^2 - 6 > 0$. Hence

$$\frac{|\mu(F_i)|}{F_i(k)} < 1 \quad (i \geq 10).$$

Expanding in the nonbacktracking basis gives

$$\int f d\mu = B_k f_0 - f(k) + \sum_{i \geq 5} f_i a_i \geq B_k f_0 - f(k).$$

Since $f \leq 0$ on $\text{supp } \mu$, $\int f d\mu \leq 0$, which proves weak duality. If equality holds, strict positivity of every high-degree slack forces $f_i = 0$ for $i \geq 5$. Equality on the positive dual support forces zeros at ξ_- , -2 , ξ_+ ; the interior zero -2 has even multiplicity because $f \leq 0$ on I_k . Degree at most four then forces f to be a scalar multiple of f_* , and $f_0 > 0$ makes the scalar positive. Equality in the graph bound would then force every nonprincipal adjacency eigenvalue to be -2 . The trace equation would give $0 = k - 2(n-1)$, or $n = (k+2)/2$, which is impossible because $n \geq k+1$. Hence the open-window bound is strict. Every symbolic identity, finite slack, tail bound, and equality-nullspace calculation is independently checked by `scripts/verify_proof_audit_02_two_sided_lp.py`. $\square$

Theorem 5.4 (Integral optimal-slack bound). *Let G be connected and k -regular with $k \geq 4$, of girth at least five and diameter three, and suppose*

$$|\theta + 1| < \sqrt{2k - 2} \quad (\theta \neq k)$$

for every adjacency eigenvalue θ . Define

$$g_k(x) = (x+2)^2((x+1)^2 - (2k-2)), \quad C_k = (k+2)^2(k^2+3), \quad h_k = 6(k+2),$$

$$\mathcal{S}_k = -g_k(A) + \frac{C_k}{n}J.$$

Then

$$\mathcal{S}_k \succeq 0, \quad \mathcal{S}_k \mathbf{1} = 0, \quad \text{tr } \mathcal{S}_k = h_k(B_k - n).$$

Moreover,

$$\mathcal{E}_k = g_k(A) - (h_k + 1)J + I$$

is a nonzero, symmetric, entrywise nonnegative integral matrix with zero diagonal and constant row sum

$$\varepsilon_{k,n} = C_k - (h_k + 1)n + 1.$$

On $\mathbf{1}^\perp$, one has $\mathcal{S}_k = I - \mathcal{E}_k$. In particular,

$$n \leq \left\lfloor \frac{(k+2)^2(k^2+3)}{6(k+2)+1} \right\rfloor.$$

Proof. The polynomial $g_k = 6(k+2)f_*$ is the optimal LP polynomial from Theorem 5.3. Thus $\mathcal{S}_k$ vanishes on $\langle \mathbf{1} \rangle$, while on a nonprincipal θ -eigenspace its eigenvalue is

$$(2k-2 - (\theta+1)^2)(\theta+2)^2 \geq 0.$$

The nonbacktracking expansion of g_k has constant coefficient h_k . Girth at least five gives $(F_i(A))_{uu} = 0$ for every u and $1 \leq i \leq 4$, and hence

$$\text{tr } \mathcal{S}_k = C_k - h_k n = h_k(B_k - n), \quad (g_k(A))_{uu} = h_k.$$

Put $z_{uv} = (g_k(A))_{uv} \in \mathbb{Z}$ for $u \neq v$, and set $a = C_k/n - h_k = (\mathcal{S}_k)_{uu}$. The 2×2 principal minor on $\{u, v\}$ gives

$$\left| \frac{C_k}{n} - z_{uv} \right| \leq a.$$

Equality on the upper side would give $(e_u - e_v)^\top \mathcal{S}_k (e_u - e_v) = 0$, hence $e_u - e_v \in \ker \mathcal{S}_k$. The strict spectral window gives

$$\ker \mathcal{S}_k = \langle \mathbf{1} \rangle \oplus E_{-2}(A).$$

Since $e_u - e_v \perp \mathbf{1}$, it would be a (-2) -eigenvector of A . Yet the u -coordinate of $A(e_u - e_v)$ is -1 if $u \sim v$ and 0 otherwise, never -2 . Therefore

$$z_{uv} \geq h_k + 1.$$

It follows directly that $\mathcal{E}_k$ has the stated entrywise properties, and $g_k(A)\mathbf{1} = C_k\mathbf{1}$ gives its row sum. The identity on $\mathbf{1}^\perp$ follows by eliminating J .

It remains to prove $\mathcal{E}_k \neq 0$. Otherwise

$$g_k(A) = (h_k + 1)J - I,$$

so $p_k(A) = 0$ on $\mathbf{1}^\perp$, where $p_k(x) = g_k(x) + 1$. After $y = x + 2$,

$$p_k(y - 2) = y^4 - 2y^3 + (3 - 2k)y^2 + 1.$$

This polynomial is irreducible over $\mathbb{Q}$. Its only possible rational roots are ± 1 , whose values are $3 - 2k$ and $7 - 2k$. A factorization into monic integer quadratics would have constant terms both 1 or both -1 ; these alternatives force the cubic and linear coefficients to be, respectively, equal or opposite, whereas they are -2 and 0 .

Rational canonical form now gives

$$\chi_{A|_{\mathbf{1}^\perp}}(x) = p_k(x)^m, \quad n - 1 = 4m.$$

The four roots of p_k sum to -6 , so $\text{tr } A = 0$ gives

$$0 = k - 6m.$$

Hence $n - 1 = 4m = 2k/3$, contradicting $n \geq k + 1$. Thus $\mathcal{E}_k \neq 0$. Its constant row sum is consequently a positive integer, so $\varepsilon_{k,n} \geq 1$, which is equivalent to the displayed order bound. The symbolic expansion, irreducibility alternatives, and finite specializations have also been checked by independent exact audits. $\square$

6. OPTIMAL-SLACK INTEGRALITY AND LOCAL POSITIVITY

The trace of $\mathcal{S}_k$ is a positive scalar multiple of the one-variable LP defect. Its 2×2 principal minors give integral local restrictions, and larger principal minors form a canonical semidefinite hierarchy. We retain the edge-local argument below because it exposes the cycle-count obstruction hidden by the stronger global order bound.

Theorem 6.1 (Three-to-one excess bound). *Under the hypotheses of Theorem 5.4, assume $k \geq 6$ and write $\varepsilon = \varepsilon_{k,n}$. The integral parameter*

$$r = 2\varepsilon - n - 2 = 2(k + 2)^2(k^2 + 3) - (12k + 27)n$$

satisfies

$$r > 0, \quad n \leq 3r.$$

Consequently

$$n \leq \left\lfloor \frac{3(k + 2)^2(k^2 + 3)}{18k + 41} \right\rfloor.$$

Proof. Put $E = \mathcal{E}_k$, $C = C_k$, $h = h_k$, and

$$\rho = \frac{\varepsilon - 1}{n} = \frac{1 + r/n}{2}.$$

The optimal slack matrix has the form

$$\mathcal{S}_k = I - E + \rho J \succeq 0.$$

We first record the divisibility identity

$$128(2C - r) = (4k + 9)(64k^3 + 112k^2 + 196k + 327) + (129 - 128r).$$

Since $n = (2C - r)/(12k + 27)$ is integral and $4k + 9$ is odd,

$$4k + 9 \mid 129 - 128r. \tag{1}$$

We shall also use the following fixed-remainder calculation. If $n = 3r + t$, then the defining equation for r gives

$$18k + 41 \mid 2C - (12k + 27)t.$$

Since $\gcd(18, 18k + 41) = 1$, Euclidean division after multiplication by 18^4 yields the necessary condition

$$18k + 41 \mid 132650 + 34992t. \tag{2}$$

Positivity of r . Suppose $r \leq 0$. A 2×2 principal minor of $\mathcal{S}_k$ gives

$$E_{uv} \leq 2 + \frac{r}{n} \quad (u \neq v).$$

Thus E is the adjacency matrix of a simple graph when $r < 0$. If $r = 0$ and $E_{uv} = 2$, positivity puts $e_u + e_v$ in $\ker S_k$. After subtracting its projection onto $\mathbf{1}$, the strict spectral window gives a (-2) -eigenvector of A . Its u -coordinate would require

$$A_{uv} = -2 + \frac{2k+4}{n},$$

which is impossible because $n \geq k^2 + 2$ and $A_{uv} \in \{0, 1\}$. Hence E is simple also when $r = 0$.

Let X be the graph with adjacency matrix $J - I - E$. It is regular of degree

$$d = n - 1 - \varepsilon = \frac{n - r - 4}{2},$$

and $A(X) = -I - E + J$ has least eigenvalue at least -2 . Every component has at least $d + 1$ vertices, so X has at most two components. We use the fact that E , being a polynomial in A and J , commutes with A ; hence every rational eigenspace of E is A -invariant. We use the classification of connected regular graphs of order greater than 28 and least eigenvalue at least -2 : such a graph is a line graph or a cocktail-party graph [14]; see also [15, Theorems 4.1.1 and 4.1.5] and [16, Introduction].

Suppose first that X is connected. The complete case would give $E = 0$, excluded in Theorem 5.4. In the cocktail-party case, E is a perfect matching and $r = -n$. The identity

$$1296C = (6k + 13)(216k^3 + 396k^2 + 654k + 1175) + 277$$

forces $k = 44$ and $n = 14812$. On the 7406-dimensional (-1) -eigenspace of E , the rational operator A is annihilated by $g_{44}(x) + 2$.

If $X = L(Y)$ and Y is q -regular, then $q \geq n/4$ and $|V(Y)| \leq 8$, contradicting $q \leq |V(Y)| - 1$ because $n \geq 38$. If Y is semiregular bipartite with part sizes $a \geq b \geq 2$, then

$$\frac{1}{a} + \frac{1}{b} = \frac{n-r}{2n} \geq \frac{1}{2}, \quad n \leq ab.$$

The cases $b \geq 3$ have $ab < 38$. For $b = 2$, connectedness forces $Y = K_{a,2}$, hence $r = -4$. Equation (1) leaves $k = 158$ and $n = 664748$. The resulting (-1) -eigenspace of E has dimension 332373.

If X has two components, write their orders as $d + 1 + a_1$ and $d + 1 + a_2$. Then

$$a_1 + a_2 = r + 2,$$

so $r \in \{-2, -1, 0\}$. Checking the divisors in (1), together with $n \geq k^2 + 2$ and kn even, leaves only

$$(r, k, n) = (-1, 62, 40875).$$

The components then have orders 20437 and 20438; one is complete and the other cocktail-party. Thus E has eigenvalue -1 with multiplicity 10219.

In all three exceptional cases the rational primary space is annihilated by $g_k(x) + 2$. This quartic is irreducible over $\mathbb{Q}$ for every integer $k \geq 6$, except $k = 7$. Indeed, after $y = x + 2$ it becomes

$$y^4 - 2y^3 + (3 - 2k)y^2 + 2.$$

Gauss's lemma reduces a quadratic factorization to monic integer quadratics whose constant terms multiply to 2; coefficient comparison leaves only $k = 4$ or $k = 7$, while the rational-root alternatives give $k = 2$ or $k = 4$. The degrees 44, 62, 158 are therefore irreducible cases, but the corresponding dimensions 7406, 10219, 332373 are not divisible by four. Since an invariant rational space annihilated by an irreducible quartic is a vector space over the degree-four field $\mathbb{Q}[x]/(g_k(x) + 2)$, its rational dimension must be divisible by four. This contradiction proves $r > 0$.

Simplicity above the putative boundary. Assume for contradiction that $n > 3r$, and put $x = r/n \in (0, 1/3)$. The 2×2 minors give $E_{uv} \leq 2$. Suppose $E_{uv} = 2$, and for $w \notin \{u, v\}$ set $s_w = E_{uw} + E_{vw}$. Cauchy-Schwarz for $e_u + e_v$ and e_w in the Gram matrix S_k gives

$$(1 + x - s_w)^2 \leq x(3 + x),$$

so the nonnegative integer s_w belongs to $\{1, 2\}$. The row sums show that exactly r vertices have $s_w = 2$. Let W be their set, let $y = \sum_{w \in W} e_w$, and put

$$e_W = \sum_{\{w,z\} \subseteq W} E_{wz}.$$

Then

$$(e_u + e_v)^T S_k (e_u + e_v) = 2x,$$

$$(e_u + e_v)^T S_k y = r(x - 1),$$

$$y^T S_k y = r + \frac{r^2(1+x)}{2} - 2e_W.$$

Positivity of this 2×2 Gram determinant yields

$$2xr + r^2(3x - 1) - 4xe_W \geq 0.$$

Since $e_W \geq 0$, it follows that $n \leq 3r + 2$. Hence $n = 3r + t$ with $t \in \{1, 2\}$. Equation (2) gives:

t	remainder	admissible $(18k + 41, k)$
1	167642	none
2	202634	$(1427, 77)$.

The remaining case gives $(k, n, r) = (77, 77831, 25943)$, but kn is odd, contrary to the handshake lemma. Therefore E is simple.

Again let X be the graph with adjacency matrix $J - I - E$. It is d -regular with least eigenvalue at least -2 . If X had at least three components, then $n \geq 3(d + 1)$, or $n \leq 3r + 6$. Thus $n = 3r + t$ for $1 \leq t \leq 6$. Equation (2) gives

t	1	2	3	4	5	6
remainder	167642	202634	237626	272618	307610	342602,

and leaves no integral graph: the $t = 2$ case is excluded above, while the sole divisor candidate for $t = 3$ does not make r integral. Hence X has at most two components.

Suppose first that X is connected. The cocktail-party case contradicts $r > 0$, so $X = L(Y)$. If Y is q -regular on v vertices, then

$$q = \frac{n - r}{4} > \frac{n}{6}, \quad v = \frac{2n}{q} < 12.$$

Simplicity and $n \geq 38$ leave only

$$(q, v; n, r) \in \{(8, 10; 40, 8), (9, 10; 45, 9), \\ (8, 11; 44, 12), (10, 11; 55, 15)\}.$$

The radius-two lower bound forces $k \leq 7$, and direct substitution in the defining formula for r excludes all four cases.

For a semiregular bipartite root with part sizes $a \geq b \geq 2$, write $p = n/a \leq q = n/b$ for its two degrees. Then

$$p + q = \frac{n - r}{2} > \frac{n}{3}, \quad (b - 2)n = b(r + 2p),$$

so $b < 6$. Connectedness excludes $p = 1$. The cases $b = 2$ and $b = 5$ are immediate; $b = 4$ leaves only

$$(p, q, a, b; n, r) = (4, 10, 10, 4; 40, 12), (4, 11, 11, 4; 44, 14),$$

both excluded by the radius-two bound and the formula for r . For $b = 3$, one has $p = 2$ or 3 , giving $n = 3r + 12$ or $n = 3r + 18$. Equation (2) gives the remainders $552554 = 2 \cdot 276277$ and $762506 = 2 \cdot 381253$. Their odd cofactors are prime and congruent to $13 \pmod{18}$, whereas $18k + 41 \equiv 5 \pmod{18}$. Thus X cannot be connected.

It remains to consider two components. For $k = 6, 7, 8$, the inequalities $r > 0$ and $n > 3r$ contain no admissible integral order. For $k \geq 9$, one has $n > 150$, and both components have order greater than 28. A regular line-graph root is too small. For a semiregular bipartite root of a component of order N , the part sizes satisfy

$$\frac{1}{a} + \frac{1}{b} = \frac{d + 2}{N} \geq \frac{n - r}{n + r + 2} > \frac{49}{100}.$$

If $b \geq 3$, the five possible pairs (a, b) have $ab \leq 18 < N$. If $b = 2$, connectedness forces the root $K_{d,2}$, whose line graph has order $2d$. Thus every component is K_{d+1} , a cocktail-party graph of order $d + 2$, or $L(K_{d,2})$. The first two types alone force $r \leq 0$. If exactly one component has order $2d$, then $n = 3r + 8$ or $n = 3r + 10$; if both do, then $n = 2r + 8$. The respective fixed remainders are

$$412586, \quad 482570, \quad 1792898.$$

The first and third have no admissible divisor of the required linear form. The second leaves only $k = 123$, for which kn is odd. This final contradiction proves $n \leq 3r$.

Substituting the definition of r gives

$$(18k + 41)n \leq 3(k + 2)^2(k^2 + 3),$$

which is the displayed bound. All polynomial divisions, irreducibility alternatives, fixed-remainder cases, line-root reductions, and Gram determinants in this argument have been replayed independently in exact arithmetic. $\square$

Corollary 6.2 (Rigidity at equality). *Under the hypotheses and notation of Theorem 6.1, equality in $n \leq 3r$ can occur only for*

$$(k, n, r) = (103, 185220, 61740).$$

This is an arithmetic parameter classification; it does not assert the existence of a graph with these parameters.

Proof. Equality gives

$$(18k + 41)r = (k + 2)^2(k^2 + 3) =: C_k.$$

Exact Euclidean division yields

$$18^4 C_k = (18k + 41)(5832k^3 + 10044k^2 + 17946k + 29107) + 66325,$$

where $66325 = 5^2 \cdot 7 \cdot 379$. Hence $18k + 41 \mid 66325$. Since $k \geq 6$, this divisor is at least 149 and is congruent to 5 modulo 18. Among the positive divisors of 66325, only 1895 has these properties. Thus $k = 103$, after which direct substitution gives $r = 61740$ and $n = 3r = 185220$. $\square$

Proposition 6.3 (Signed-complement bridge). *Under the hypotheses and notation of Theorem 6.1, assume $0 < r < n$ and define*

$$S = (6k + 14)J - 2I - g_k(A).$$

Then S is a signed adjacency matrix:

$$S_{uu} = 0, \quad S_{uv} \in \{-1, 0, 1\} \quad (u \neq v).$$

It has constant signed row sum $(n - r - 4)/2$, and

$$S + 2I \succeq 0, \quad E_{-2}(S) = E_{-2}(A).$$

Proof. Write $x = r/n \in (0, 1)$, $\rho = (1 + x)/2$, and $E = \mathcal{E}_k$. Since

$$S_k = I - E + \rho J \succeq 0,$$

its 2×2 principal minors give $-1 \leq E_{uv} \leq 2 + x$. The entries of E are nonnegative integers, so $E_{uv} \in \{0, 1, 2\}$ for $u \neq v$, and $S = J - I - E$ has the asserted entries. The row sum follows from $\varepsilon = (n + r + 2)/2$. Moreover,

$$S + 2I = S_k + \frac{n - r}{2n}J \succeq 0.$$

The matrices A and S commute. On a nonprincipal adjacency eigenvector with eigenvalue θ , the corresponding eigenvalue of S is $-2 - g_k(\theta)$. The only zero of g_k in the open shifted WOW window is $\theta = -2$; the other two zeros are its excluded endpoints. The principal eigenvalue of S is not -2 because $r < n$. Hence the two (-2) -eigenspaces coincide. $\square$

Proposition 6.4 (Edge-local cycle bounds). *Let G be connected and k -regular, where $k \geq 4$, of girth at least five and diameter three, and suppose*

$$|\theta + 1| \leq \sqrt{2k - 2} \quad (\theta \neq k)$$

for every adjacency eigenvalue θ . For an edge uv , let σ_{uv} be the number of 5-cycles containing that edge. Then

$$2k - 2 \leq \sigma_{uv} \leq \frac{2(k + 2)^2(k^2 + 3)}{n} - 10k - 26.$$

If $n = k^2 + 1 + c$, then also

$$\sigma_{uv} \geq (k - 1)^2 - c.$$

Proof. Recall g_k and $C_k = g_k(k)$ from Theorem 5.4, and put

$$M = -g_k(A) + \frac{C_k}{n}J.$$

The spectral window gives $M \succeq 0$. For an edge uv ,

$$(A^3)_{uv} = \sum_{z \sim v} (A^2)_{uz} = k + (k - 1) = 2k - 1.$$

Here $z = u$ contributes k , while every other neighbour of v is at distance two from u and has a unique length-two path from u . Moreover,

$$(A^4)_{uv} = \sum_z (A^2)_{uz} (A^2)_{zv} = \sigma_{uv}.$$

Indeed, the nonzero summands away from the diagonal are precisely the vertices at distance two from both u and v . Their two unique length-two paths, together with uv , form a five-cycle, and each five-cycle through uv yields one such vertex. Since $C_k = (k + 2)^2(k^2 + 3)$, the diagonal and edge entries of M are

$$a = \frac{C_k}{n} - 6(k + 2), \quad b = \frac{C_k}{n} - (4k + 14) - \sigma_{uv}.$$

The principal submatrix on $\{u, v\}$ is $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$, so $a \geq 0$ and $-a \leq b \leq a$. The inequality $b \leq a$ gives $\sigma_{uv} \geq 2k - 2$, while $b \geq -a$ gives the stated upper bound.

For the final bound, every radius-two ball has size $k^2 + 1$. The set

$$\{u, v\} \cup (N(u) \setminus \{v\}) \cup (N(v) \setminus \{u\})$$

contains $2k$ vertices and lies in $B_2(u) \cap B_2(v)$. Every further intersection vertex is at distance two from both endpoints and is therefore in the preceding five-cycle bijection. Hence

$$|B_2(u) \cap B_2(v)| = 2k + \sigma_{uv}.$$

Inclusion–exclusion and $n = k^2 + 1 + c$ now give $\sigma_{uv} \geq (k - 1)^2 - c$. The complete walk classification, sign directions, and radius-two bijection have also been checked independently and exactly. $\square$

For $\ell \geq 3$, let N_ℓ denote the number of ℓ -cycles in G .

Corollary 6.5 (Edge–cycle divisibility sieve). *Under the hypotheses of Proposition 6.4, put*

$$L_{k,n} = \max\{2k - 2, 2k^2 - 2k + 2 - n\}, \quad U_{k,n} = \frac{2(k+2)^2(k^2+3)}{n} - 10k - 26.$$

Necessarily

$$\lceil L_{k,n} \rceil \leq \lfloor U_{k,n} \rfloor.$$

If both sides equal an integer s , then every edge of G lies in exactly s five-cycles and

$$5 \mid \frac{skn}{2}.$$

Proof. Writing $n = k^2 + 1 + c$, the two lower bounds in Proposition 6.4 combine to

$$\sigma_{uv} \geq \max\{2k - 2, (k - 1)^2 - c\} = L_{k,n},$$

while its upper bound is $U_{k,n}$. Since σ_{uv} is an integer, the first conclusion follows. If the two integer bounds coincide at s , then $\sigma_{uv} = s$ for every edge. Counting edge–five-cycle incidences gives

$$5N_5 = \sum_{uv \in E(G)} \sigma_{uv} = s|E(G)| = \frac{skn}{2},$$

which proves the divisibility condition. $\square$

Theorem 6.6. *Every connected 6-regular strict counterexample to WOW-284 has order at most 50.*

Proof. A separate Rayleigh and trace argument shows that any degree-six strict counterexample has diameter three. For vertices at distance $d \geq 4$, the vector with weights 3, 1, -3 , -1 on the two endpoints and their respective neighbourhoods has Rayleigh quotient at most

$$\frac{204 - 81d}{15} \leq -8.$$

Thus $\lambda_{\min}(D(G)) \leq -8$, so $\Phi(G) \leq 6 - 8 = -2$, contradicting strictness. Diameter two would force equality in the Moore bound, hence order 37 and adjacency characteristic polynomial $(x - 6)(x^2 + x - 5)^{18}$; its root sum is -12 , contradicting $\text{tr } A = 0$. Diameter one is excluded by the girth hypothesis. Thus the diameter is three, and Theorem 6.1 gives

$$n \leq \left\lfloor \frac{3 \cdot 8^2 \cdot 39}{149} \right\rfloor = 50.$$

For an independent local explanation of the excluded boundary, assume $n = 51$. Corollary 6.5 has $\lceil L_{6,51} \rceil = \lfloor U_{6,51} \rfloor = 11$, and hence requires

$$5N_5 = 153 \cdot 11 = 1683, \quad 5 \nmid 1683,$$

which is impossible. An independent exact audit verifies the full diameter reduction and incidence calculation. $\square$

Corollary 6.7 (Low-degree order windows). *There is no regular strict counterexample of degree at most five. In degrees six through nine, every regular strict counterexample satisfies*

$$\begin{aligned} k = 6 : & \quad n \leq 50, \\ k = 7 : & \quad n = 50 \text{ in diameter two, or } n \leq 74 \text{ in diameter three,} \\ k = 8 : & \quad n \leq 108, \\ k = 9 : & \quad n \leq 150. \end{aligned}$$

Proof. The degree exclusion is Theorem 4.2, Theorem 4.3 excludes diameter at least five, and Theorem 4.4 removes diameter four for $k \leq 9$. For diameter three, Theorem 6.1 gives, for $k = 6, 7, 8, 9$, respectively,

$$n \leq 50, 75, 108, 150.$$

Since $7n = 2|E(G)|$, the degree-seven bound improves to 74. The diameter-two alternatives are determined by the Moore multiplicities: only the degree-seven, order-50 Hoffman–Singleton case survives. $\square$

For comparison, the unadjusted diameter-three bounds in degrees 6 through 20 are

k	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
n	50	75	108	150	201	263	336	422	521	635	765	911	1075	1257	1459.

When k is odd, $kn = 2|E(G)|$ requires n to be even; in particular, the entries for $k = 7, 11, 15, 17, 19$ improve to 74, 262, 634, 910, 1256, respectively.

Proposition 6.8 (The order-50 (-2) -multiplicity bound). *Let G be 6-regular, of order 50, and of girth at least five. The multiplicity $m_{-2}(A)$ of the adjacency eigenvalue -2 satisfies*

$$m_{-2}(A) \leq 20.$$

Proof. Put $m = m_{-2}(A)$, remove the principal eigenvalue 6 and the m copies of -2 , and denote the remaining spectral moments by $\mu_0, \dots, \mu_4$. Girth at least five gives

$$\text{tr } A = 0, \quad \text{tr } A^2 = 300, \quad \text{tr } A^3 = 0, \quad \text{tr } A^4 = 3300,$$

and therefore

$$(\mu_0, \mu_1, \mu_2, \mu_3, \mu_4) = (49 - m, -6 + 2m, 264 - 4m, -216 + 8m, 2004 - 16m).$$

The moment matrix

$$H = \begin{pmatrix} \mu_0 & \mu_1 & \mu_2 \\ \mu_1 & \mu_2 & \mu_3 \\ \mu_2 & \mu_3 & \mu_4 \end{pmatrix}$$

is positive semidefinite. Exact expansion gives

$$\det H = 3600(1625 - 81m) \geq 0.$$

Since m is an integer, $m \leq 20$. $\square$

6.1. Necessary structure at order fifty. The remaining degree-six boundary is highly constrained, although not yet eliminated.

Theorem 6.9. *Let G be a connected 6-regular graph of order 50 and girth at least five, and suppose $\Phi(G) > 0$. Then G has diameter three. For an edge e , let σ_e be the number of five-cycles containing e . Then $\sigma_e \in \{12, 13\}$. Call e low or high according as $\sigma_e = 12$ or 13, respectively, and let H be the spanning subgraph of high edges. Put $m = |E(H)|$, and let $\tau(v)$ be the number of five-cycles through v . Then*

$$\begin{aligned} \tau(v) &\in \{36, 37, 38\}, & d_H(v) &= 2\tau(v) - 72 \in \{0, 2, 4\}, \\ m &\equiv 0 \pmod{5}, & N_5 &= 360 + \frac{m}{5}. \end{aligned}$$

For a two-edge path $u - v - w$, put

$$R_{uvw} = 6\alpha_{uvw} + \beta_{uvw},$$

where α and β count the five- and six-cycles containing the path. The allowed values are

types of the two incident edges	R_{uvw}
low–low	30, 31, 32
mixed	30, 31, 32
high–high	30, 31.

Writing $S_2 = \sum_v d_H(v)^2$, one has

$$\begin{aligned} 1950 - m \leq N_6 &\leq 2200 - \frac{5m}{6} - \frac{S_2}{12}, \\ \frac{43m^2 - 70200m + 119632500}{58500} &\leq N_6 \leq \frac{4220000 - 2200m - 7m^2}{2000}. \end{aligned}$$

Writing

$$n_i = |\{v \in V(G) : d_H(v) = i\}| \quad (i \in \{0, 2, 4\}),$$

exact enumeration leaves 266 triples (n_0, n_2, n_4) such that

$$n_0 + n_2 + n_4 = 50, \quad m = n_2 + 2n_4 \equiv 0 \pmod{5},$$

and at least one integer N_6 satisfies all four displayed bounds.

Proof. The diameter reduction used in Theorem 6.6 applies verbatim: diameter at least four gives a Rayleigh quotient at most -8 , and diameter two gives the trace contradiction from $(x - 6)(x^2 + x - 5)^{18}$. Hence the diameter is three. Proposition 6.4 gives

$$L_{6,50} = 12, \quad U_{6,50} = \frac{346}{25} < 14.$$

Since σ_e is integral, $\sigma_e \in \{12, 13\}$ for every edge e . Around a fixed vertex v , write $\Gamma_i = \Gamma_i(v)$ for the distance layers; their sizes are 1, 6, 30, 13. If τ is the number of five-cycles through v , the average row quotient is similar to the symmetric compression on normalized layer indicators. Its nonprincipal factor q_τ satisfies

$$195q_\tau(-1 + \sqrt{10}) = (-215 + 56\sqrt{10})\tau + 7350 - 1860\sqrt{10}.$$

The coefficient of τ is negative because $56^2 \cdot 10 < 215^2$, and at $\tau = 39$ the right-hand side is $9(-115 + 36\sqrt{10}) < 0$. Also

$$195q_\tau(6) = 1500(75 - \tau).$$

Every vertex of Γ_2 has exactly one neighbour in Γ_1 , and the edges inside Γ_2 are in bijection with the five-cycles through v . Hence

$$e(\Gamma_2, \Gamma_3) = 30 \cdot 5 - 2e(\Gamma_2) = 150 - 2\tau > 0,$$

because every vertex of the nonempty layer Γ_3 has a neighbour in Γ_2 . Thus $q_\tau(6) > 0$. Therefore $\tau \geq 39$ would place a nonprincipal compression eigenvalue in $(-1 + \sqrt{10}, 6)$, contradicting interlacing and the open WOW window. Conversely,

$$150 - 2\tau = e(\Gamma_2, \Gamma_3) \leq 13 \cdot 6,$$

so $\tau \geq 36$. Hence $\tau \in \{36, 37, 38\}$. Finally, $\sum_{e \ni v} \sigma_e = 2\tau(v)$, so the high-edge degree is $2\tau(v) - 72$; counting edge-five-cycle incidences gives $5N_5 = 12 \cdot 150 + m$.

For a two-path $u - v - w$, girth at least five gives $(A^3)_{uw} = \alpha_{uvw}$ and $(A^4)_{uw} = 16 + \beta_{uvw}$. The corresponding 3×3 principal minor of the centered positive-semidefinite matrix yields the displayed finite sets for R_{uvw} , except for an apparent equality value $R_{uvw} = 29$. At equality, the Gram norm of $e_u - e_w$ vanishes, so this vector lies in the kernel of the centered matrix. On $\mathbf{1}^\perp$, that matrix is $-g_6(A)$. The strict shifted window excludes the two endpoint zeros of g_6 , leaving only its double zero at -2 ; hence the kernel there is exactly the adjacency -2 -eigenspace. However, the u -coordinate of $A(e_u - e_w)$ is zero because $u \not\sim w$, whereas the u -coordinate of $-2(e_u - e_w)$ is -2 . This excludes $R_{uvw} = 29$. Summing the local inequalities gives the first pair of N_6 bounds; shifted moment and localizing matrices give the second. The complete symbolic determinants, Schur complements, kernel argument, and exact profile enumeration are independently checked by `scripts/verify_proof_audit_06_order50_feasibility.py`. The surviving 266 profiles are exact pruning data, not an existence or nonexistence claim. $\square$

Remark 6.10 (Signed-root Gram formulation). Under the hypotheses of Theorem 6.9, put

$$T = 50J - g_6(A) - 2I.$$

Then

$$T\mathbf{1} = 2\mathbf{1}, \quad T + 2I \succeq 0, \quad 25S_6 = 25(T + 2I) - 2J.$$

This is the specialization of Proposition 6.3: the excess parameter is $r = 42$, so the signed row sum is 2. For vertices u, z at distance three, put

$$q_{uz} = 6(A^3)_{uz} + (A^4)_{uz}.$$

The 2×2 estimate gives $q_{uz} \in \{48, 49, 50, 51\}$, and the kernel argument in Theorem 5.4 excludes 48. Together with the edge and two-path intervals in Theorem 6.9, this gives

$$T_{uv} \in \{-1, 0, 1\} \quad (u \neq v), \quad T_{uu} = 0.$$

Consequently $T + 2I$ is the Gram matrix of 50 vectors of norm $\sqrt{2}$ with pairwise inner products in $\{-1, 0, 1\}$ and constant signed row sum 2. Its kernel is $E_{-2}(A)$, so Proposition 6.8 gives

$$\text{rank}(T + 2I) \geq 30.$$

Thus the unresolved degree-six boundary can be viewed as a root-type integral Gram problem constrained by the graph's distance classes and local cycle data.

Theorem 6.11 (Disconnected signed complement). *Under the hypotheses of Theorem 6.9, the edge-signed graph with adjacency matrix T from Remark 6.10 is disconnected.*

Proof. Let N_i denote the number of i -cycles in G . In terms of the degree-six nonbacktracking polynomials,

$$(g_6 + 2)^2 = 28144F_0 + 18220F_1 + 8838F_2 + 3576F_3 + 1233F_4 \\ + 352F_5 + 78F_6 + 12F_7 + F_8.$$

Girth at least five gives $\text{tr } F_i(A) = 0$ for $1 \leq i \leq 4$. For lengths five and six, every closed nonbacktracking walk is a directed cycle. At lengths seven and eight there are also the walks obtained by attaching a one-edge tail to a directed five- or six-cycle. Hence

$$\begin{aligned} \text{tr } F_5(A) &= 10N_5, & \text{tr } F_6(A) &= 12N_6, \\ \text{tr } F_7(A) &= 14N_7 + 40N_5, & \text{tr } F_8(A) &= 16N_8 + 48N_6. \end{aligned}$$

Removing the principal adjacency contribution and restoring the principal signed eigenvalue 2 now gives

$$\text{tr } T^2 = 8(500N_5 + 123N_6 + 21N_7 + 2N_8 - 604100).$$

If P_+ and P_- are the numbers of positive and negative signed edges, then the signed row sum and the zero diagonal give

$$P_+ - P_- = 50, \quad \text{tr } T^2 = 2(P_+ + P_-) = 100 + 4P_-.$$

It follows that P_- is odd.

Suppose that T were connected. By Proposition 6.8, $\text{rank}(T + 2I) \geq 30$. The root-system representation theorem for connected edge-signed graphs with smallest eigenvalue at least -2 [17, Theorem 2] shows that $T + 2I$ admits a representation in a root system of type D_ℓ or E_8 . The rank bound excludes E_8 . Thus

$$B^T B = T + 2I$$

for an integral matrix B whose columns b_u are roots $\pm e_i \pm e_j$. Put $s = B\mathbf{1}$. Since $(T + 2I)\mathbf{1} = 4\mathbf{1}$,

$$b_u \cdot s = 4 \quad \text{for every } u, \quad \|s\|^2 = 200.$$

Changing signs of coordinate axes if necessary, assume $s_i \geq 0$. The coordinate-support multigraph is connected, since otherwise the columns of B would split into two Gram-orthogonal families and T would be disconnected. If v coordinates are used, then

$$30 \leq v \leq 51 :$$

the lower bound is the Gram rank, and a connected support multigraph with fifty root-edges has at most fifty-one vertices.

For every root, two coordinate levels satisfy $\pm s_i \pm s_j = 4$. Connectivity therefore places all levels in one of

$$0, 4, 8, \dots; \quad 2, 6, 10, \dots; \quad 1, 3, 5, \dots$$

The first family would make $\|s\|^2$ divisible by 16, contrary to $200 \equiv 8 \pmod{16}$.

In the second family, a level at least 10, together with the other $v - 1 \geq 29$ levels, would contribute at least $10^2 + 29 \cdot 2^2 > 200$. Thus only levels 2 and 6 occur. If their multiplicities are n_2, n_6 , then

$$n_2 + 9n_6 = 50, \quad v = n_2 + n_6 = 50 - 8n_6 \geq 30,$$

so

$$(n_2, n_6) \in \{(50, 0), (41, 1), (32, 2)\}.$$

Every root is now either $e_i + e_j$ with $s_i = s_j = 2$, or $e_i - e_j$ with $s_i = 6$ and $s_j = 2$, after ordering the two coordinates. Its sign pattern is unique on a fixed support, while duplicate columns would have inner product 2; hence distinct roots share at most one coordinate in this restricted family. At a level-two coordinate let p_i, m_i be its positive and negative incidence counts. Then $p_i - m_i = 2$, and every level-six coordinate has six incidences, each paired with a negative incidence at a level-two coordinate. A negative Gram product occurs precisely when two roots meet at one level-two coordinate with opposite incidence signs, so it is counted exactly once in

$$P_- = \sum_{i:s_i=2} p_i m_i.$$

Consequently

$$P_- \equiv \sum_{i:s_i=2} p_i m_i \equiv \sum_{i:s_i=2} m_i = 6n_6 \equiv 0 \pmod{2},$$

contradicting the parity already proved.

It remains to exclude odd levels. Let a_t, c_t count coordinates at levels $4t + 1, 4t + 3$, respectively. Flow balance along the two level chains gives the following identity. Difference roots cancel within their chain when signed incidences are summed, while the sum roots $e_i + e_j$ between levels 1 and 3, the only cross-chain type, contribute equally to both sides:

$$\sum_{t \geq 0} (4t + 1)a_t = \sum_{t \geq 0} (4t + 3)c_t.$$

Using this identity with $\|s\|^2 = 200$ yields

$$\frac{200 - 3v}{32} = \sum_{t \geq 1} \binom{t+1}{2} (a_t + c_t).$$

Thus $v \equiv 24 \pmod{32}$, which is impossible for $30 \leq v \leq 51$. All three level families are excluded, proving that T is disconnected. $\square$

7. DISTANCE SPECTRA OF PUNCTURED MOORE GRAPHS

Let M be a degree- k Moore graph of diameter two and put $\Delta = \sqrt{4k - 3}$.

Theorem 7.1 (One deleted vertex). *For $v \in V(M)$, put $H = M - v$. Then*

$$|V(H)| = k^2, \quad \delta^*(H) = k - \frac{1}{k}, \quad \lambda_{\min}(D(H)) = -2 - \sqrt{k}.$$

The complete distance spectrum is

$$\begin{aligned} \text{Spec } D(H) = & \{\rho_+, \rho_-\} \cup \{(-2 + \sqrt{k})^{(k-1)}, (-2 - \sqrt{k})^{(k-1)}\} \\ & \cup \left\{ \left(-\frac{\Delta + 3}{2} \right)^{(m_+)}, \left(\frac{\Delta - 3}{2} \right)^{(m_-)} \right\}, \end{aligned}$$

where

$$\begin{aligned} \rho_{\pm} &= k^2 - 2 \pm \sqrt{k(k^3 - 2k^2 + 3k - 1)}, \\ m_{\pm} &= \frac{k(k-2)(\Delta \pm 1)}{2\Delta}. \end{aligned}$$

Thus

$$\Phi(H) = k - \frac{1}{k} - 2 - \sqrt{k},$$

which is positive exactly for integers $k \geq 5$.

Theorem 7.2 (Endpoints of an edge). *Let $uv \in E(M)$ and $H = M - \{u, v\}$. For $k \geq 3$,*

$$|V(H)| = k^2 - 1, \quad \delta^*(H) = k - \frac{2}{k}, \quad \lambda_{\min}(D(H)) = -2 - \sqrt{k}.$$

The complete distance spectrum is

$$\begin{aligned} \text{Spec } D(H) = & \{\sigma_+, \sigma_-, k - 4\} \\ & \cup \{(-2 + \sqrt{k})^{(2k-4)}, (-2 - \sqrt{k})^{(2k-4)}\} \\ & \cup \left\{ \left(-\frac{\Delta + 3}{2} \right)^{(a_+)}, \left(\frac{\Delta - 3}{2} \right)^{(a_-)} \right\}, \end{aligned}$$

where

$$\begin{aligned} \sigma_{\pm} &= k^2 - 3 \pm \sqrt{k^4 - 2k^3 + 3k^2 - 8k + 7}, \\ a_+ &= \frac{(k-2)(k + (k-2)\Delta)}{2\Delta}, \quad a_- = \frac{(k-2)((k-2)\Delta - k)}{2\Delta}. \end{aligned}$$

Thus

$$\Phi(H) = k - \frac{2}{k} - 2 - \sqrt{k},$$

which is positive exactly for integers $k \geq 5$.

Proof architecture for Theorems 7.1 and 7.2. The displayed dual-degree formulas are the cases $s = 1$ and $s = 2$ of Theorem 8.1; its proof below is independent of the spectral decompositions. For one deleted vertex, put $A = N(v)$, $B = \Gamma_2(v)$, let C be the A -by- B incidence matrix, and let B_0 be the adjacency matrix on B . Only pairs inside A lose their unique length-two path. For distinct $a, a' \in A$, choose a neighbour $b \in B$ of a . The vertices b, a' are nonadjacent, and their unique common neighbour c lies in B : it is not v , since $b \not\sim v$, and it is not in A , since an edge inside A would form a triangle through v . Hence $a - b - c - a'$ is a surviving path, and therefore

$$D(H) = \begin{pmatrix} 3(J - I) & 2J - C \\ 2J - C^T & 2(J - I) - B_0 \end{pmatrix}.$$

The normalized constant quotient is

$$Q_1 = \begin{pmatrix} 3(k-1) & (2k-1)\sqrt{k-1} \\ (2k-1)\sqrt{k-1} & 2k^2 - 3k - 1 \end{pmatrix},$$

whose eigenvalues are $\rho_{\pm}$. Block comparison in the Moore identity gives

$$\begin{aligned} CC^{\top} &= (k-1)I, & CB_0 &= J - C, & C^{\top}\mathbf{1}_A &= \mathbf{1}_B, \\ B_0^2 + C^{\top}C &= (k-1)I - B_0 + J. \end{aligned}$$

Thus $C^{\top}$ is injective and

$$\mathbb{R}^B = \langle \mathbf{1}_B \rangle \perp C^{\top}(\mathbf{1}_A^{\perp}) \perp K, \quad K = \ker C, \quad \dim K = k(k-2).$$

Choose an orthogonal basis $\mathcal{B}$ of $\mathbf{1}_A^{\perp}$, and for $z \in \mathcal{B}$ put

$$W_z = \text{span}\{(z, 0), (0, C^{\top}z)\}.$$

The cell-constant space W_{const} , the two-dimensional modules W_z for $z \in \mathcal{B}$, and $0 \oplus K$ are mutually orthogonal and invariant. Put

$$W_{\text{inc}} = \bigoplus_{z \in \mathcal{B}} W_z.$$

Consequently

$$\mathbb{R}^{V(H)} = W_{\text{const}} \perp W_{\text{inc}} \perp (0 \oplus K), \quad 2 + 2(k-1) + k(k-2) = k^2.$$

On each zero-sum incidence module the action matrix in these generators is $\begin{pmatrix} -3 & -(k-1) \\ -1 & -1 \end{pmatrix}$, giving $-2 \pm \sqrt{k}$. On K , $B_0^2 + B_0 - (k-1)I = 0$, $D = -2I - B_0$, and the trace of B_0 is zero after removing the principal eigenvalue $k-1$ and the $k-1$ copies of -1 on $C^{\top}(\mathbf{1}_A^{\perp})$. Dimension and trace therefore give $m_{\pm}$.

For an adjacent deleted pair, put

$$A = N(u) \setminus \{v\}, \quad B = N(v) \setminus \{u\}, \quad C = V(M) \setminus (\{u, v\} \cup A \cup B).$$

Only pairs inside A or inside B lose a length-two path. For distinct $a, a' \in A$, choose a residual neighbour c of a . The vertices c, a' are nonadjacent, and their unique common neighbour is also residual: it cannot be one of the deleted endpoints, cannot lie in A by triangle-freeness, and cannot lie in B because no edge joins A to B . This gives a surviving length-three path; the argument for B is symmetric. The antisymmetric constant line has eigenvalue $k-4$, and the normalized symmetric quotient is

$$Q_2 = \left(\frac{5k-8}{\sqrt{(k-1)(2k-3)(4k-6)}} \quad \frac{\sqrt{(k-1)(2k-3)(4k-6)}}{2k^2-5k+2} \right),$$

with eigenvalues $\sigma_{\pm}$. Let R_A, R_B be the incidence matrices from A, B to C , and let T be the adjacency matrix on C . The Moore identity gives

$$\begin{aligned} R_A R_A^{\top} &= R_B R_B^{\top} = (k-1)I, & R_A R_B^{\top} &= J, \\ R_A T &= J - R_A, & R_B T &= J - R_B, \\ R_A^{\top} R_A + R_B^{\top} R_B + T^2 &= (k-1)I - T + J. \end{aligned}$$

The zero-sum row images of R_A and R_B are injective and orthogonal. Each combines with its cell zero-sum space to form an invariant module carrying $\begin{pmatrix} -3 & -(k-1) \\ -1 & -1 \end{pmatrix}$. The residual space

$$K = \ker R_A \cap \ker R_B, \quad \dim K = (k-2)^2,$$

is invariant and satisfies

$$T^2 + T - (k-1)I = 0, \quad D = -2I - T.$$

These spaces and the three-dimensional cell-constant space are mutually orthogonal and complete, since

$$3 + 2(2k-4) + (k-2)^2 = k^2 - 1.$$

The constant direction of T has eigenvalue $k-2$, while its two zero-sum incidence images contribute trace $-2(k-2)$. Since $\text{tr } T = 0$, one has $\text{tr}(T|_K) = k-2$, and dimension and trace give $a_{\pm}$. The residual negative root is greater than $-2 - \sqrt{k}$. For $k \geq 3$, the antisymmetric constant root also satisfies

$$(k-4) - (-2 - \sqrt{k}) = k-2 + \sqrt{k} > 0.$$

The same is true of the smaller constant-quotient roots: in each case the leading diagonal entry of the shifted quotient is positive, and

$$\begin{aligned} \det(Q_1 + (2 + \sqrt{k})I) &= k(2k^2 + 2k^{3/2} - 3k + 2) > 0, \\ \det(Q_2 + (2 + \sqrt{k})I) &= (\sqrt{k}-1)(\sqrt{k}+1)(2k^2 + 2k^{3/2} - 3k + 2\sqrt{k} + 6) > 0. \end{aligned}$$

Exact quotient normalization, trace-to-multiplicity equations, replacement paths, and all least-root comparisons are independently checked by `scripts/verify_proof_audit_05_small_moore_punctures.py`. $\square$

Theorem 7.3 (Two nonadjacent deleted vertices). *Let u, v be nonadjacent vertices of M , $k \geq 5$, and put $H = M - \{u, v\}$. Define*

$$\begin{aligned} R_k(x) &= x^4 + (10 - 2k^2)x^3 + (2k^3 - 17k^2 - 2k + 36)x^2 \\ &\quad + (12k^3 - 49k^2 - 4k + 53)x \\ &\quad - 2k^4 + 17k^3 - 38k^2 + 5k + 20, \\ M_- &= \frac{k(k-2) + (k^2 - 4k + 2)\Delta}{2\Delta}, \quad M_+ = \frac{-k(k-2) + (k^2 - 4k + 2)\Delta}{2\Delta}. \end{aligned}$$

Then

$$\begin{aligned} \chi_{D(H)}(x) &= (x - k + 3)R_k(x)(x^2 + 4x - k + 3)^{k-2} \\ &\quad \cdot (x^2 + 4x - k + 5)^{k-2} \left(x + \frac{\Delta + 3}{2}\right)^{M_-} \\ &\quad \cdot \left(x - \frac{\Delta - 3}{2}\right)^{M_+}. \end{aligned}$$

Moreover,

$$\delta^*(H) = k - \frac{2}{k},$$

and H is a strict counterexample for every realizable integer $k \geq 6$.

Proof. The deleted vertices have a unique common neighbour w . With $A = N(u) \setminus \{w\}$, $B = N(v) \setminus \{w\}$, $C = N(w) \setminus \{u, v\}$, and

$$Z = V(M) \setminus (\{u, v, w\} \cup A \cup B \cup C),$$

the surviving graph has cell sizes, in the order $\{w\}, A, B, C, Z$,

$$1, \quad k-1, \quad k-1, \quad k-2, \quad (k-1)(k-2).$$

The Moore common-neighbour rule makes the A – B edges a perfect matching and the five-cell partition equitable. It also supplies a length-three replacement path for every pair whose unique length-two path used u or v ; hence every such new distance is exactly three. On the cell-constant space, the row-sum distance quotient is

$$\begin{pmatrix} 0 & 3k-3 & 3k-3 & k-2 & 2(k-1)(k-2) \\ 3 & 3k-6 & 2k-3 & 2k-4 & 2k^2-7k+6 \\ 3 & 2k-3 & 3k-6 & 2k-4 & 2k^2-7k+6 \\ 1 & 2k-2 & 2k-2 & 2k-6 & 2k^2-7k+5 \\ 2 & 2k-3 & 2k-3 & 2k-5 & 2k^2-7k+5 \end{pmatrix}.$$

Its characteristic polynomial is $(x - k + 3)R_k(x)$.

Identify $\mathbb{R}^A$ with $\mathbb{R}^B$ through the perfect matching, and let R_A, R_B, R_C be the incidence matrices from the three nonconstant cells to Z . Let T be the adjacency matrix on Z . The Moore identity gives

$$\begin{aligned} R_A R_A^\top &= R_B R_B^\top = (k-2)I, \quad R_C R_C^\top = (k-1)I, \\ R_A R_B^\top &= J - I, \quad R_A R_C^\top = R_B R_C^\top = J, \\ R_A T + R_B &= J - R_A, \quad R_B T + R_A = J - R_B, \quad R_C T = J - R_C, \\ R_A^\top R_A + R_B^\top R_B + R_C^\top R_C + T^2 &= (k-1)I - T + J. \end{aligned}$$

Moreover, every vertex of Z has one neighbour in each of A, B, C , so

$$R_A^\top \mathbf{1}_A = R_B^\top \mathbf{1}_B = R_C^\top \mathbf{1}_C = \mathbf{1}_Z, \quad T \mathbf{1}_Z = (k-3)\mathbf{1}_Z.$$

Use the coordinate order $(\{w\}, A, B, C, Z)$. Choose orthogonal bases $\mathcal{B}_A$ of $\mathbf{1}_A^\perp$ and $\mathcal{B}_C$ of $\mathbf{1}_C^\perp$. For $\xi \in \mathcal{B}_A$ and $\eta \in \mathcal{B}_C$, put

$$\begin{aligned} u_\xi^\pm &= (0, \xi, \pm\xi, 0, 0), \quad v_\xi^\pm = (0, 0, 0, 0, (R_A^\top \pm R_B^\top)\xi), \\ u_\eta^C &= (0, 0, 0, \eta, 0), \quad v_\eta^C = (0, 0, 0, 0, R_C^\top \eta), \end{aligned}$$

and define

$$W_\xi^\pm = \text{span}\{u_\xi^\pm, v_\xi^\pm\}, \quad W_\eta^C = \text{span}\{u_\eta^C, v_\eta^C\}.$$

The displayed identities show that the relevant incidence maps are injective, that these two-dimensional modules are mutually orthogonal, and that they are orthogonal to the five-dimensional cell-constant space W_{const} . For

$$K = \ker R_A \cap \ker R_B \cap \ker R_C$$

one has

$$K \subseteq \mathbf{1}_Z^\perp, \quad \dim K = (k-2)(k-4), \quad T(K) \subseteq K.$$

Thus W_{const} , the modules $W_\xi^\pm$, W_η^C , and $0 \oplus 0 \oplus 0 \oplus 0 \oplus K$ form an orthogonal direct sum, since

$$5 + 4(k-2) + 2(k-3) + (k-2)(k-4) = k^2 - 1.$$

In the ordered generator pairs (u_ξ^+, v_ξ^+) , (u_ξ^-, v_ξ^-) , and (u_η^C, v_η^C) , respectively, the distance operator has matrices

$$\begin{pmatrix} -4 & -(k-3) \\ -1 & 0 \end{pmatrix}, \quad \begin{pmatrix} -2 & -(k-1) \\ -1 & -2 \end{pmatrix}, \quad \begin{pmatrix} -2 & -(k-1) \\ -1 & -1 \end{pmatrix}.$$

These give the two displayed quadratic factors and $k-3$ copies of each Moore linear factor. On K , the Z -block identity gives

$$T^2 + T - (k-1)I = 0, \quad D(H)|_K = -2I - T.$$

The constant direction of Z has T -eigenvalue $k-3$, while the Z -images in the symmetric, antisymmetric, and common-neighbour modules have eigenvalues $-2, 0, -1$, with dimensions $k-2, k-2, k-3$, respectively. Since $\text{tr } T = 0$,

$$\text{tr}(T|_K) = 2(k-2).$$

Dimension and trace give the residual multiplicities; after adding the common-neighbour copies, they are precisely M_- and M_+ .

The value $\delta^*(H) = k - \frac{2}{k}$ is again the $s = 2$ case of Theorem 8.1. For strictness, put

$$E_S = D(H) - D(M)[V(H)].$$

Up to a simultaneous permutation of rows and columns, E_S is the adjacency matrix of two copies of K_k meeting in w , together with $k(k-2)$ isolated vertices. Hence

$$\lambda_{\min}(E_S) = \frac{k-2 - \sqrt{k^2 + 4k - 4}}{2}.$$

Since the parent Moore graph has score $k - (3 + \Delta)/2$ and deletion lowers the minimum dual degree by $2/k$, Proposition 7.4 gives

$$\Phi(H) \geq \frac{3k-5-\Delta - \sqrt{k^2 + 4k - 4}}{2} - \frac{2}{k}.$$

For $k \geq 6$,

$$\Delta < 2\sqrt{k} - \frac{3}{4\sqrt{k}}, \quad \sqrt{k^2 + 4k - 4} < k + 2 - \frac{4}{k+2}.$$

The lower bound is therefore greater than

$$f(k) = k - \frac{7}{2} - \sqrt{k} + \frac{3}{8\sqrt{k}} + \frac{2}{k+2} - \frac{2}{k}.$$

Now $f(6) = 29/12 - 15\sqrt{6}/16 > 0$, since $116^2 > 6 \cdot 45^2$. Moreover $f'(x) > 0$ for $x \geq 6$: the absolute values of the three negative terms in $f'(x)$ have sum less than $1/4 + 1/64 + 1/32 < 1$, while $2/x^2 > 0$. Thus $\Phi(H) > 0$. The direct-sum decomposition, injectivity, orthogonality, all recomputed distances, and the strict comparison are independently checked by `scripts/verify_proof_audit_03_nonadjacent_puncture.py` and `scripts/verify_research_extensions_exact.py`. $\square$

Proposition 7.4 (Deletion stability). *Let G be connected with at least two vertices, let $S \subseteq V(G)$, and suppose $H = G - S$ is connected with at least two vertices. Put*

$$D_0 = D(G)[V(H)], \quad E_S = D(H) - D_0, \\ a = \delta^*(G), \quad b = \delta^*(H), \quad \gamma = \Phi(G).$$

Then

$$\boxed{\Phi(H) \geq \gamma - (a - b) + \lambda_{\min}(E_S)}.$$

Proof. One has

$$D(H) + bI = (D_0 + aI) + E_S - (a - b)I.$$

Principal-submatrix interlacing gives $\lambda_{\min}(D_0 + aI) \geq \gamma$, and Weyl's inequality gives the result. For Moore punctures, E_S is an explicit distance-increase matrix; the exact specializations are checked by `scripts/verify_research_extensions_exact.py`. $\square$

Theorem 8.1 (Small-puncture normal form). *Let M be a degree- k Moore graph of diameter two, let $S \subseteq V(M)$ have size $s \leq k - 1$, and put $H = M - S$. Then H is connected, has diameter at most three, and*

$$\delta^*(H) = k - \frac{s}{k}.$$

Let B be the surviving-vertex by deleted-vertex incidence matrix, with $B_{xz} = 1$ exactly when $x \sim_M z$. Then

$$D(H) = 2(J - I) - A(H) + BB^\top - \text{diag}(BB^\top).$$

Proof. The case $s = 0$ is immediate, so assume $1 \leq s \leq k - 1$. Let $x, y \in V(H)$ be nonadjacent in M , and suppose their unique common neighbour z lies in S . For every $a \in N_M(x) \setminus \{z\}$, the vertices a, y are nonadjacent and have a unique common neighbour b_a . Thus

$$x - a - b_a - y$$

is a length-three path. These $k - 1$ paths are internally vertex-disjoint: a shared vertex $b_a = b_{a'}$ would give a four-cycle, while $b_a = a'$ would give a triangle through x . Besides z , at most $s - 1 \leq k - 2$ vertices are deleted, so one path survives. Hence a destroyed length-two path becomes distance exactly three, which proves the matrix formula.

For $x \in V(H)$, let $t_x = |N_M(x) \cap S|$. A deleted neighbour of x is adjacent to no surviving neighbour of x , by triangle-freeness; each other deleted vertex has at most one common neighbour with x . Thus

$$\sum_{y \in N_H(x)} t_y \leq s - t_x.$$

Consequently,

$$d_H^*(x) \geq k - \frac{s - t_x}{k - t_x} \geq k - \frac{s}{k},$$

where the last inequality follows from

$$\left(k - \frac{s - t}{k - t}\right) - \left(k - \frac{s}{k}\right) = \frac{t(k - s)}{k(k - t)} \geq 0.$$

The intersection bound

$$\left| \bigcap_{z \in S} \Gamma_2(z) \right| \geq k^2 + 1 - s(k + 1) \geq 2$$

provides a surviving vertex x at distance two from every deleted vertex. For $z \in S$, let y_z be the unique common neighbour of x, z . If $y_z \in S$, then the choice of x would require $d_M(x, y_z) = 2$, contradicting $x \sim y_z$. Thus all witnesses survive, each deleted vertex contributes exactly once to the neighbour-degree deficit, and equality is attained. The replacement paths, boundary case $s = k - 1$, distance formula, and attainment step are independently checked by `scripts/verify_proof_audit_12_small_puncture.py`. $\square$

Corollary 8.2 (Uniform deletion stability). *Under the hypotheses of Theorem 8.1, put*

$$t_x = |N_M(x) \cap S|, \quad \tau(S) = \max_{x \in V(M-S)} t_x.$$

Then

$$\Phi(M - S) \geq k - \frac{3 + \sqrt{4k - 3}}{2} - \frac{s}{k} - \tau(S).$$

In particular, the deletion is a strict counterexample whenever the right-hand side is positive. Since $\tau(S) \leq s$, every deletion of s vertices is strict whenever

$$s \left(1 + \frac{1}{k}\right) < k - \frac{3 + \sqrt{4k - 3}}{2}.$$

For $k > 3$, equivalently, every deletion of at most

$$r_k = \min \left\{ k - 1, \left\lceil \frac{k}{k + 1} \left(k - \frac{3 + \sqrt{4k - 3}}{2} \right) \right\rceil - 1 \right\}$$

vertices is strict. In particular, $r_7 = 2$; for a hypothetical degree-57 Moore graph, whose existence remains open [18], one would have $r_{57} = 47$.

Proof. By Theorem 8.1, the distance-increase matrix is

$$E_S = BB^\top - \text{diag}(t_x).$$

Since $BB^\top \succeq 0$ and $\text{diag}(t_x) \preceq \tau(S)I$, one has $\lambda_{\min}(E_S) \geq -\tau(S)$. The parent Moore graph has score

$$\Phi(M) = k - \frac{3 + \sqrt{4k - 3}}{2},$$

and Theorem 8.1 shows that deletion lowers the minimum dual degree by s/k . Proposition 7.4 now gives the first bound. The second follows from $\tau(S) \leq s$, and solving its strict scalar inequality for the largest integral s gives r_k . $\square$

Theorem 8.3 (Hoffman–Singleton robustness radius). *Let M be the Hoffman–Singleton graph. Every induced graph $M - S$ with $|S| \leq 5$ is a strict counterexample to WOW-284. This is sharp in the universal sense: there exists a six-vertex set whose deletion is not strict. Hence the universal vertex-deletion robustness radius is exactly five.*

Proof. Theorem 8.1 gives

$$\delta^*(M - S) = \frac{49 - |S|}{7}.$$

Corollary 8.2 already proves strictness uniformly for $|S| \leq 2$. To obtain the sharp universal radius, two explicitly stored permutations are verified edge-by-edge as automorphisms of the coordinate graph; the group they generate has order 252000. Their orbits on deletion sets of sizes 0, 1, 2, 3, 4, 5 have counts

$$1, 1, 2, 4, 11, 33.$$

For every one of the 52 representatives, exact fraction arithmetic gives

$$7D(M - S) + (49 - |S|)I \succ 0.$$

Orbit-size sums equal $\binom{50}{s}$, so every labelled set is covered.

For sharpness, delete

$$\{P_{2,4}, P_{3,1}, P_{3,4}, Q_{2,1}, Q_{3,4}, Q_{4,4}\}.$$

The resulting graph has $\delta^* = 43/7$, while an exact $LDL^\top$ decomposition of $7D + 43I$ has exactly one negative and no zero pivot. Thus it is not strict. Generator action, orbit exhaustion, BFS distances, the small-puncture formula, and a handwritten rational $LDL^\top$ implementation are independently checked by `scripts/verify_proof_audit_13_hs_robustness.py`. $\square$

Remark 8.4. The theorem asserts that every deletion through size five succeeds and that at least one deletion of size six fails. It does not assert that every six-vertex deletion fails.

9. EQUALITY AND OBSTRUCTIONS TO NATURAL CONSTRUCTION FAMILIES

Theorem 9.1 (Equality boundary). *Let G be connected, k -regular, of girth at least five and diameter three. Then*

$$\Phi(G) = 0 \iff \max_{\theta \neq k} |\theta + 1| = \sqrt{2k - 2}.$$

Equivalently, $D + kI$ is positive semidefinite and singular. If $2k - 2$ is not a square, the two boundary adjacency eigenvalues occur with equal multiplicity, so the distance eigenvalue $-k$ has even multiplicity. If $2k - 2$ is a square, then $k = 2r^2 + 1$ for some integer r .

Proof. This is immediate from Theorem 3.1. In the nonsquare case, the two boundary values are algebraic conjugates, so their multiplicities in the integral characteristic polynomial agree. In the square case, $2k - 2 = s^2$ forces s even. The exact scalar audit is `scripts/verify_equality_boundary.py`. $\square$

Jørgensen’s 9-regular order-96 graph of girth five is an exact equality case:

$$\delta^* = 9, \quad \lambda_{\min}(D) = -9,$$

with multiplicity eight. The construction is due to Jørgensen [19]; the three local graph representations, handwritten graph6 decoder, characteristic polynomials, root intervals, and provenance boundary are checked by `scripts/verify_proof_audit_09_jorgensen96.py`.

9.1. A prime-field obstruction. For an odd prime $q \geq 7$ and $1 \leq m \leq q$, define $G(q, m)$ on vertices $P_{i,j}, Q_{k,\ell}$, with $0 \leq i, k < m$ and $j, \ell \in \mathbb{F}_q$, by

$$\begin{aligned} P_{i,j} &\sim P_{i,j\pm 1}, \\ Q_{k,\ell} &\sim Q_{k,\ell\pm 2}, \\ P_{i,j} &\sim Q_{k,ik+j}. \end{aligned}$$

This is a balanced specialization of known finite-field girth-five constructions [20]. It is $(m+2)$ -regular. A coordinate common-neighbour calculation shows that it has no triangle or 4-cycle for $q \geq 7$: for a cross pair, the possible same-side common neighbours require residues in the disjoint sets $\{\pm 1\}$ and $\{\pm 2\}$.

Theorem 9.2. *If $G(q, m)$ has diameter three, then it is not a strict counterexample to WOW-284.*

Proof. Theorem 4.2 excludes $m \leq 3$. The zero Fourier block has eigenvalues $m+2$, $2-m$, and 2 with multiplicity $2m-2$; the shifted WOW window leaves only $m \in \{4, 5, 6\}$. Let $\omega = e^{2\pi i/q}$. On the nonzero character $t = 1$, the adjacency block has form

$$\begin{pmatrix} aI & M \\ M^* & bI \end{pmatrix}, \quad a = 2 \cos \frac{2\pi}{q}, \quad b = 2 \cos \frac{4\pi}{q}, \quad M_{ik} = \omega^{ik}.$$

If σ is a singular value of M , the associated two-dimensional invariant subspace carries $\begin{pmatrix} a & \sigma \\ \sigma^* & b \end{pmatrix}$. Since $\|M\|_F^2 = m^2$, one singular value satisfies $\sigma^2 \geq m$, and the block has a nonprincipal eigenvalue at least

$$\sqrt{m} + \frac{a+b}{2} \geq \sqrt{m} + \cos(\pi/7) - \frac{1}{2}.$$

For $q \geq 7$, both $\cos(2\pi/q)$ and $\cos(4\pi/q)$ increase with q , so their sum is minimized at $q = 7$. The last inequality then uses $2 \cos(2\pi/7) + 2 \cos(4\pi/7) = 2 \cos(\pi/7) - 1$. Moreover $\cos(\pi/7) > \sqrt{3}/2$, and the increasing function $h(m) = \sqrt{2m+2} - \sqrt{m}$ satisfies

$$h(m) \leq h(6) = \sqrt{14} - \sqrt{6} < \frac{27}{20} < \frac{\sqrt{3}+1}{2}.$$

Thus the displayed nonprincipal eigenvalue lies above $-1 + \sqrt{2m+2}$ for $m = 4, 5, 6$. The common-neighbour case split, Fourier reduction, radical comparisons, and exact $q = 7$ controls are independently checked by `scripts/verify_proof_audit_08_prime_field.py`. $\square$

9.2. Layer-respecting matching deletions. For $\pi \in S_5$, delete the perfect matching

$$\mathcal{M}_\pi = \{ \{P_{i,j}, Q_{\pi(i), i\pi(i)+j}\} : i, j \in \mathbb{F}_5 \}$$

from the Hoffman–Singleton graph.

Theorem 9.3. *Each deletion produces a connected simple 6-regular graph of order 50, girth five, and diameter four. The 120 labelled graphs form exactly two isomorphism classes. The 20 affine permutations have*

$$\lambda_{\min}(D) = -13, \quad \Phi = -7,$$

and the 100 nonaffine permutations have

$$\lambda_{\min}(D) = -6 - \sqrt{61}, \quad \Phi = -\sqrt{61}.$$

Thus none of these graphs is a counterexample.

Proof. Every $P_{i,j}$ occurs once in $\mathcal{M}_\pi$; for a fixed $Q_{k,\ell}$, the unique incident matching edge is obtained from $i = \pi^{-1}(k)$ and $j = \ell - ik$. Thus $\mathcal{M}_\pi$ is a perfect matching, and its deletion leaves a simple 6-regular graph. Deleting edges cannot create a short cycle, while the same-layer pentagons remain; exact breadth-first search gives connectedness and diameter four.

Explicit type-preserving and type-swapping coordinate automorphisms generate orbits of sizes 20 and 100. The representatives have different adjacency characteristic polynomials, so the orbits are distinct isomorphism classes. Exact distance characteristic polynomials and Sturm separators give the stated least roots. All 120 matchings, 400 coordinate maps, 48,000 matching images, orbit coverage, graph hypotheses, and root certificates are checked by `scripts/verify_proof_audit_07_layer_matchings.py`. $\square$

10. EXACT COMPUTATION AND FORMAL VERIFICATION

The analytic arguments above are primary. Exact computation is used in three roles: to certify explicitly labelled finite graphs, to exhaust precisely specified finite orbit families, and to check symbolic identities whose derivations are supplied. No theorem-level sign or eigenvalue ordering uses floating-point arithmetic. The exact computations use SymPy and NetworkX [21, 22].

The accompanying release archives the labelled graph data, exact rational and polynomial certificates, and finite-orbit records used here, so the computer-assisted steps are reproducible without a file-by-file index in the paper.

The explicit 50-vertex counterexample is fully formalized in Lean 4.31 with Mathlib 4.31 [23, 24]. The development pins the exact toolchain and dependency revision in `lean/lean-toolchain` and `lean/lake-manifest.json`. It checks the coordinate graph, regularity, the exhaustive common-neighbour certificate, girth five, the adjacency-square and distance-matrix identities, and an exact rational diagonalization with multiplicities. Thus the result is verified at graph level, including its least distance eigenvalue and strict WOW-284 gap.

Lean also kernel-checks finite spectral certificates attached to the explicit constructions of orders 38, 39, 40, 42. At orders 38, 39, 42, they certify the dual-degree data and positive definiteness of the relevant shifted distance matrix. At order 40, they certify an invertible exact diagonalization, multiplicities, least eigenvalue -5 , dual degree six, and gap one. These are finite matrix certificates rather than end-to-end `SimpleGraph.dist` formalizations.

Separately, Lean formalizes the analytic optimization statement in Theorem 5.3. For every integer $k \geq 4$ and every admissible finitely supported expansion

$$f = \sum_i c_i F_i, \quad c_0 > 0, \quad c_i \geq 0 \quad (i \geq 5), \quad f|_{I_k} \leq 0,$$

it proves

$$B_k c_0 \leq f(k), \quad B_k = \frac{(k+2)(k^2+3)}{6}.$$

The formal development defines the coefficient family of the displayed quartic f_* , proves that it is admissible and attains equality, and proves that the equality cases are exactly the positive scalar multiples of f_* , both as polynomials and at coefficient level. The public Lean development is sorry-free and kernel-checked by Lean 4.31.

This LP formalization is deliberately graph-independent. It does not formalize the trace interpretation of the $F_i(A)$, the girth-five vanishing and nonnegativity statements, or the passage from the LP inequality to graph-order bounds. The integral-slack consequences, signed-complement theory, puncture spectra, and deletion results lie outside this Lean development. They are proved in the text using the analytic and, where explicitly identified, exact computer-assisted components. This is the precise scope of the Lean claims in the paper.

11. BROADER MATHEMATICAL QUESTIONS

The mechanisms in this paper point beyond the conjecture that motivated them. They connect extremal polynomial certificates, local positive-semidefinite constraints, perturbations of metric operators, exact computation, and formal proof. At that methodological level, they suggest the following questions.

- (1) Given an optimal polynomial or semidefinite certificate for a global spectral inequality, when do the principal minors of its slack operator form a complete hierarchy of local constraints? Can rank or flat-extension conditions force finite convergence and reconstruct the extremal discrete objects?
- (2) Is there a general stability theory for spectral inequalities in which a small global defect forces proximity to a structured algebraic model, and local perturbations distinguish sporadic objects from finite shadows of infinite families?
- (3) Which one-variable spectral arguments survive when homogeneity or commutativity is lost? Can degree-weighted operators, nonbacktracking operators, or matrix-valued orthogonal polynomials provide comparably sharp certificates for irregular or multitype systems?

OpenAI ChatGPT-5.6 Sol Pro assisted with adversarial proof checking, proof exploration, and Lean formalization. The author assumes full responsibility for the mathematics, attribution, and conclusions. The source, exact certificates, and build instructions are available at github.com/SamPetkov/wow284 and correspond to release v2.2.7.

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